assignments in

Junior Science

book 6

Motion

CHRIS GREEF
SUADA BILALI

Quality ready-to-use resources

assignments in

Junior Science

book 6

Motion

CHRIS GREEF
SUADA BILALI

INSTANT LESSONS

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Commissioning editor: Jennifer Gough
Managing editor: Ruth Schultz
Editor: Michael Quinn
Cover design: Gail McManus Graphics
Printed in Australia by Ligare Book Printer

Blake EDUCATION
Better ways to learn

Instant Lessons
An imprint of Blake Education Pty Ltd
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108 Main Road
Clayton South VIC 3169
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www.blake.com.au

ISBN 978 1 92209 081 2

Introduction

Each book in the *Assignments in Junior Science* series provides a selection of worksheets covering one or two major topics. The worksheets are self-contained, supplying the information necessary for students to complete the given tasks.

As students work through these activities, they will interpret and analyse data including text, tables, graphs and diagrams; and organise and represent data in a variety of formats. They will also utilise basic scientific skills such as working with formulas, completing calculations, making generalisations and explaining what they have discovered. The activities incorporate a range of theoretical and practical scientific skills, and also involve the development of literacy skills. Sample answers are provided at the back of the book.

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Measuring speeds – a 50 metre track

1

Speed up your calculations by using a calculator.

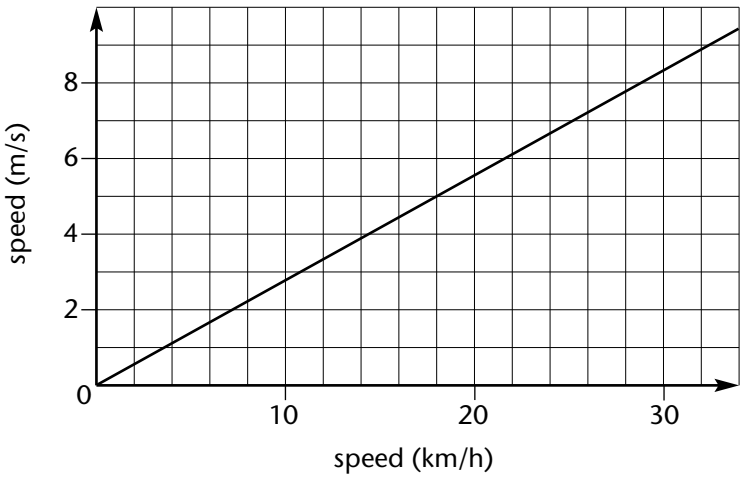
After beginning to learn about speed and velocity, our class went outside for an experiment. We worked with five students in each group.

One student in each group used a trundle wheel to mark a distance of 50 metres in the sportsground. The 50 metres became the group race track. The start and finish points were clearly marked. In each group, one person with a stopwatch stood next to the finish line and another member of the group stood at the start. A third member of the group was selected to walk, another to jog and the fifth one to run. The timekeeper standing at the finish line started the stopwatch as soon as the signal from the starter was given. The stopwatch was stopped when the walker, jogger or runner crossed the finish line.

This event, from starting point to finish line, was repeated five times (each of the five students played each role). The measurements were recorded in the table below.

Group results	Walking (s)	Jogging (s)	Running (s)
Student 1	33.3	22.2	7.5
Student 2	32.0	25.3	7.0
Student 3	35.4	23.0	8.0
Student 4	28.3	26.4	9.5
Student 5	30.0	24.1	8.5
Average time calculation			
Average time answer			
Average speed calculation			
Average speed (m/s) answer			
Average speed (km/h) answer			

Speed conversion graph



1 Calculate the average group time for each method of covering the 50 metres. Record the answers in the cell spaces provided.

2 Calculate the average group speed for walking, jogging and running. (speed = distance/time, that is, v = d/t) Record your answer in metres per second.

3 Use the speed conversion graph to convert these speeds to kilometres per hour.

4 In your science book, write up this experiment as a scientific record. Use the subheadings: Aim, Equipment, Method, Safety, Results, Conclusion. You are not required to repeat the results table exactly, but include a table of the results which you think are needed.

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How do speeds compare?

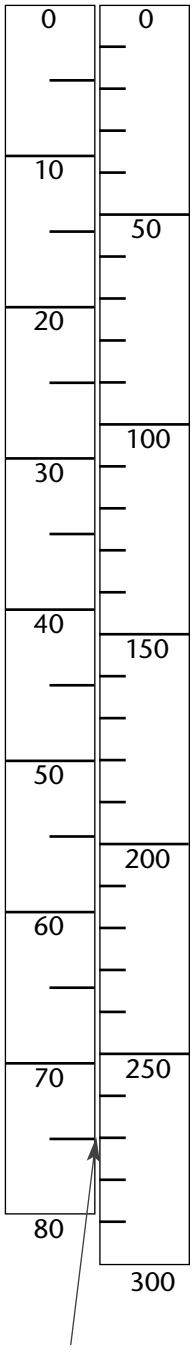
2a

Speed up your calculations by using a calculator.

A champion adult runner covered 100 m in 10 s, which gives his average speed as 10 m/s. (Average speed = 100/10 = 10 m/s) A leopard runs at a speed of 80 km/h for a short time. Who is the faster runner?

The speed converter column and the speed converter graph below relate speeds measured in km/h to the same speeds measured in m/s. This means that they help you to convert between speeds in m/s and km/h.

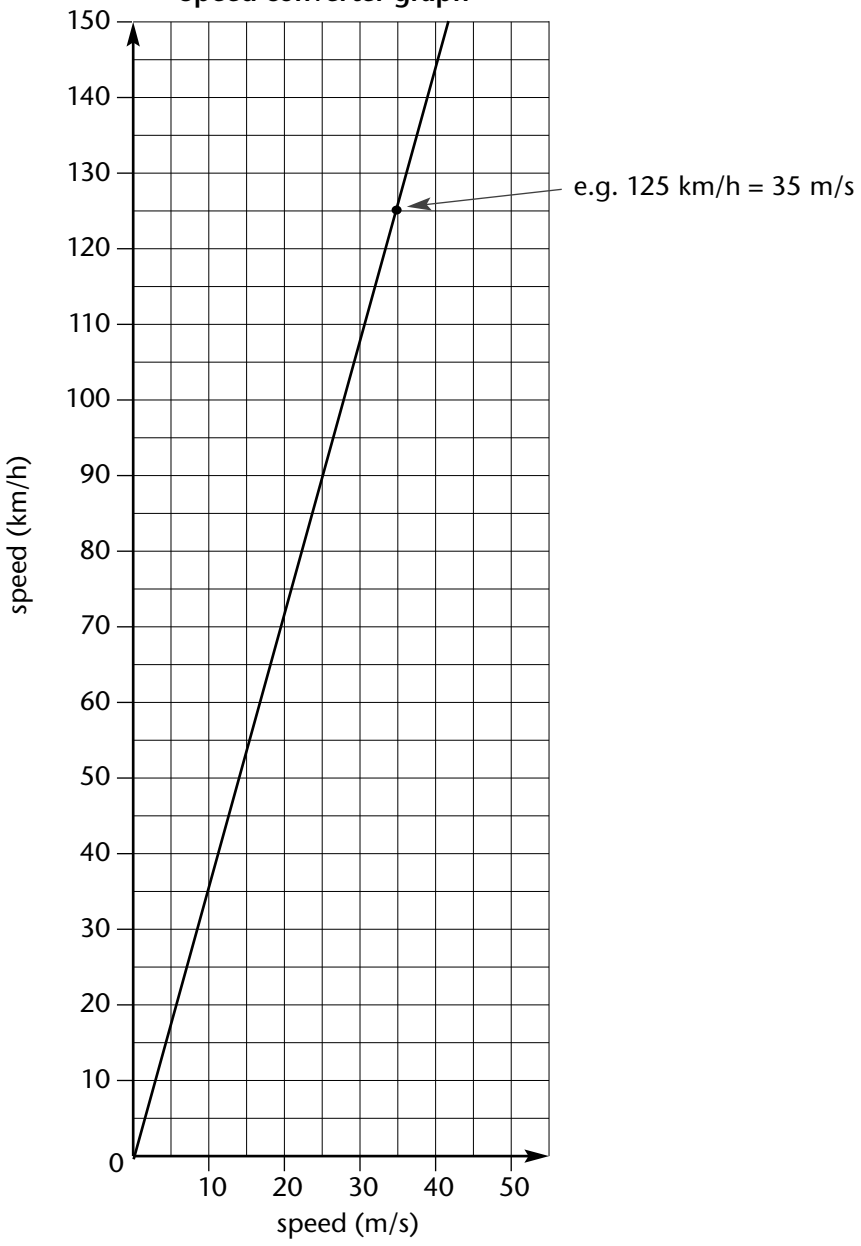
Speed converter column



1 Complete the following table using the speed converter column.

Speed in km/h	Speed in m/s
0 (waiting at the traffic lights)	
	10 (running really, really fast)
80 (a leopard at full speed)	
	1 (walking slowly)
60 (a car driving at town speed limit)	
	70 (speed of a golf ball)

Speed converter graph



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How do speeds compare? 2b

2 Complete the following table using the speed converter graph on the previous page.

Speed in km/h	Speed in m/s
40 (speed limit near a school)	
	6 (running a 400 m race)
110 (speed limit on highways)	
	2 (a fly in full flight)
20 (riding a pushbike)	
	20 (ball speed after a 7 storey drop)

You can convert measurements in one unit to measurements in another unit more accurately by using a calculator. In order to change a speed of 1 metre per second to kilometres per hour, the metre is replaced by 1/1000 km (there are 1000 m in 1 km, so there is 1/1000 km in 1 m) and the second is replaced by 1/3600 of an hour (there are 60 × 60 = 3600 seconds in 1 hour, so there is 1/3600 h in 1 s).

So: 1 m/s = 1 × 1/1000 km /1/3600 h = 1 × 3600/ 1000 km/h = 3.6 km/h

Multiplying the magnitude of a m/s speed by a factor of 3.6 will give the magnitude of the equivalent speed in km/h.

Similarly, to change a speed of 1 kilometre per hour to a speed written in metres per second, parallel substitutions are made:

1 km/h = 1 × 1000 m/3600 s = 1 × 1000/3600 m/s = 1/3.6 m/s = 0.28 m/s

Multiplying the magnitude of a km/h speed by a factor of 0.28 will give the magnitude of the equivalent speed in m/s. Alternatively, you can divide the km/h speed by 3.6 to find the equivalent m/s speed.

3 Calculate equivalent speeds for each speed mentioned in the table. Record the calculations and the speeds calculated in the appropriate cells in the table below.

Speed in km/h	Speed in m/s
1000 (speed of a jet aeroplane)	
	300 (speed of sound)
100 (average speed in quarter mile race)	
	300 000 000 (speed of light)
372 (Formula 1 race track speed record)	
	1500 (speed of a bullet)

4 Calculate the km/h equivalents to the following m/s speeds. Compare your calculated values with those given in the speed converter column on the previous page.

m/s	Calculation	Calculated speed km/h	Converter column km/h
0			
10			
20			
30			
40			
50			
80			

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Formulas for energy and motion

3a

Speed up your calculations by using a calculator.

1 There are a number of formulas that are used when calculating things like speed, distance and acceleration. Let's see how good you are with some of those formulas. Connect the matching items (quantity, symbol, formula, unit, comment) in the table below by drawing lines between them.

Quantity	Symbol	Formula	Unit	Comment
time	a	d/t	J	acceleration due to earth's gravity
speed	d	d/v	J	amount of matter
potential energy	F	F/a	kg	change in speed per unit time
mass	g	ma	m	energy due to height
kinetic energy	h	mgh	m	energy due to movement
height	KE	$^1/_2\,mv^2$	m/s	height above sea level
gravity acceleration	m	vt	m/s^2	how far
force	PE	$(v_2 - v_1)/t$	m/s^2	how fast
distance	v	PE/(mg)	N	how long
acceleration	t	PE/(mh)	s	strength

Note: potential energy = gravitational energy and kinetic energy = movement energy

2 Use formulas from question 1 to answer the following questions. Write the correct formula, show your working and give your answer as a number with units.

a It takes a sports car 7 s to reach a speed of 100 km/h (= 28 m/s). What is the car's acceleration?

b The sports car has a mass of one tonne. How much kinetic energy would it possess at 100 km/h (= 28 m/s), at 50 km/h (= 14 m/s), at 25 km/h (= 7 m/s) and at 0 km/h?

c The sports car speeds along at 100 km/h for 5 s. What distance does it cover? Use the m/s speed.

d The car travels another 200 m at that speed. How long does it take to cover the 200 m?

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The electromagnetic spectrum 3b

e If it continued straight ahead, the car would leave the road and plummet 200 m downwards. Assume that the acceleration due to gravity is 10 m/s^2. Find the car's potential energy relative to the bottom of the 200 m fall.

f The car, travelling at 100 km/h, brakes hard and comes to a stop in 4 s. Assuming constant deceleration, what is the deceleration during the stopping time? Use the m/s speed.

g Another similar car, travelling at the same speed, does not fare as well. It hits a wall at 100 km/h and comes to a fairly sudden stop in half a second (fortunately the car has crumple zones which stop the car more slowly). What is its acceleration? Use m and s.

3 A cyclist of mass 50 kg rides a pushbike of mass 15 kg. The cyclist is riding at a speed of 72 km/h (= 20 m/s).

a What is the kinetic energy of the cyclist and bike?

b What distance does the cyclist cover in 30 s?

c At a constant speed of 72 km/h how long would it take the cyclist to ride 1 km?

d The cyclist brakes and stops in 4 s. What is the cyclist's deceleration?

e Assume that the force of the cycle tyres on the road is uniform during the stopping period. What force is acting to stop the cyclist?

f Imagine that this cyclist and bike were to fall off a building. How high would the building have to be in order to provide the bike and cyclist with the same amount of energy that they had when travelling at 72 km/h?

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The convenient unit

4a

Speed up your calculations by using a calculator.

Physical measurements such as 100 km/h have two parts; a number (the size or measure, e.g. 100) and a unit (e.g. km/h). Today the units most scientists use are from the metric system.

1 Complete the table below by writing the correct unit beside each quantity, and then writing the abbreviation of each unit. The first one has been done for you.
Choose from these units: ampere, byte, joule, litre, kilogram, metre, newton, second, volt.

Quantity	Base unit name	Base unit abbreviation
Distance	*metre*	*m*
Time		
Mass		
Volume		
Weight		
Energy		
Electric current		
Voltage (potential difference)		
Computer memory size		

In order to keep the number a meaningful and friendly size, prefixes are often used as part of the unit. The 'magic prefix numbers' in the metric system are 1000 (10^3) and 0.001 (10^{-3}). Kilo is the prefix indicating a multiplier of 1000. Milli is the prefix meaning 1/1000.

1000 millilitres make up 1 litre; and a millilitre equals 1/1000 litre or 0.001 litre.

2 **a** Complete the table which shows the value of a prefix and its abbreviation.

Name	Value: number	Value: exponent	Abbreviation
milli		10^{-3}	
micro	1/1 000 000		
nano		10^{-9}	
pico	1/1 000 000 000 000		p
kilo		10^{3}	
mega	1 000 000		M
giga		10^{9}	

b Prefix abbreviations are listed in the first column of the table. Complete the table below. Provide examples using these prefixes in relation to the metre. The first one has been done for you.

Prefix	Example: name	Example with units abbreviation	Fraction of a metre	Decimal of a metre	Magnitude in index form	Number to make one metre
m	millimetre	1 mm	1/1000 m	0.001 m	10^{-3}	1000 mm
µ						
n						
k						
M						
G						

Note that centimetres (1 cm = 0.01 m; 100 cm = 1 m) and decimetres (1 dm = 0.1 m; 10 dm = 1 m) are used in some situations; however, metres and millimetres are preferred by scientists, architects, builders and trades people.

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The convenient unit

4b

3 People choose to use a prefix so that the number part of the measurement is convenient and meaningful. Saying that a plane trip is 3 000 000 m long is awkward. Saying 3 000 km or 3 Mm makes the expression of the distance easier to understand. Choose a better unit in which to express the measurements listed in the following table.

Distance	Measurement	Better unit
Length of a football field	100 m	
Length of a marathon running track	40 000 m	
Diameter of a tiny dot of ink	0.001 m	
Diameter of a pencil dot	0.000 1 m	
Distance from earth to moon	300 000 000 m	
Distance between earth and sun	150 000 000 000 m	
Thickness of an onion cell	0.000 05 m	
Length of a virus	0.000 000 2 m	

4 **a** Measuring time is a little more complicated. It is common to use prefixes with seconds (s) – but only when the time is less than a second. For larger time measurements we switch to minutes (min) and hours (h) and then to days (days) and years (y). Complete the table below about time measurements.

Unit name	Unit symbol	Equivalent in seconds	Number per second
millisecond			1 000 ms
microsecond		1/1000 000	
minute	min		
hour			
day			
year			

b Express the time measurements for each event in the table in a more meaningful unit.

Event	Time	Standard measurement
Light travelling from sun to earth	480 s	
Sound travelling from Sydney to Canberra	1 020 s	
Car driving from Melbourne to Canberra	21 600 s	
Student doing serious science homework	1 800 s	
Driver having a micro sleep	0.001 s	
Electric energy moving from power point to television	0.000 000 01 s	
Rock falling off a cliff	0.002 h	
Time for a dog to eat its dinner	0.01 h	

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How long will it take?

5a

Speed up your calculations by using a calculator.

The table shows sample speeds for a variety of activities.

Activity	Speed (km/h)	Speed (m/s)
Boing 747 plane flying at 10 km altitude	900	
Space shuttle revolving Earth in orbit	25 000	
Truck travelling at speed limit	100	
Car driving at town speed limit	60	
Car travelling at suburban speed limit	50	
Car racing during the Bathurst 1000	150	
Cheetah running at full speed	80	
Person walking	5	
Person jogging	10	
Marathon runner racing	20	
Short distance runner racing	10	
Cyclist pedaling along	25	
Tour de France cyclist racing	40	

When you want to change a speed from kilometres per hour to metres per second, you substitute 1000 metres for the kilometre and 3600 seconds for the hour.

Let's convert the speed 2 km/h to a speed in m/s:

2 km/h = 2 . 1000m/3600s = 2 . 1000/3600 m/s = 2 . 1/3.6 m/s = 0.56 m/s

So, walking slowly and covering 2 km in 1 h is walking at a speed of 0.56 m in 1 s.

1 Follow the reverse of the process above to convert the speed of 2 m/s to km/h. (Remember that 1 m equals 1/1000 km and that 1 s is 1/3600 h.)

In summary:

- To convert km/h to m/s, divide the km/h speed by 3.6 (? km/h = ?/3.6 m/s)
- To convert m/s to km/h, multiply the m/s speed by 3.6 (? m/s = ? . 3.6 km/h)

2 Calculate the m/s values for the activities listed in the table at the top of this page, and record them in the third column.

The table below shows the sizes of five distances.

Distance	Size in km	Size in m
The distance across a sports field		100
The length of a Sydney Harbour Bridge span		500
Sydney Harbour Tunnel length	2.3	
Melbourne to Canberra distance	1000	
Canberra to Sydney distanced	300	

3 Calculate and record the missing measurements in the table above.

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How long will it take?

5b

4 State a formula for calculating **speed**, given the **distance** travelled and the **time** taken to travel that distance. Use the bold terms.

5 Rearrange this formula for speed to make time the subject.

6 Select five different speeds from the first table on the previous page. Write them down the left side of the chart below. Now select three different distances from the second table on the previous page. Write them across the top row. Calculate how long it would take to travel each distance at each speed.

		Distances		
Speeds				

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Speed and things

6a

The boxes below contain notes related to motion. Use information from these notes to help in answering the following questions.

Deceleration is negative acceleration. It occurs when an unbalanced force acts on an object to oppose its movement.

Acceleration is the change in velocity and can be calculated using the equation

$$a = (v_2 - v_1)/t$$

where v_2 = final velocity, v_1 = initial velocity and t = time taken.

Common units for speed are km/h or m/s.

Velocity is the measure of distance travelled in a unit time. The concept involves direction as well.

Speed is the distance travelled in a unit time.
speed = distance/time

Distance-time graphs can be used to find speeds and speed changes as well as distances travelled.

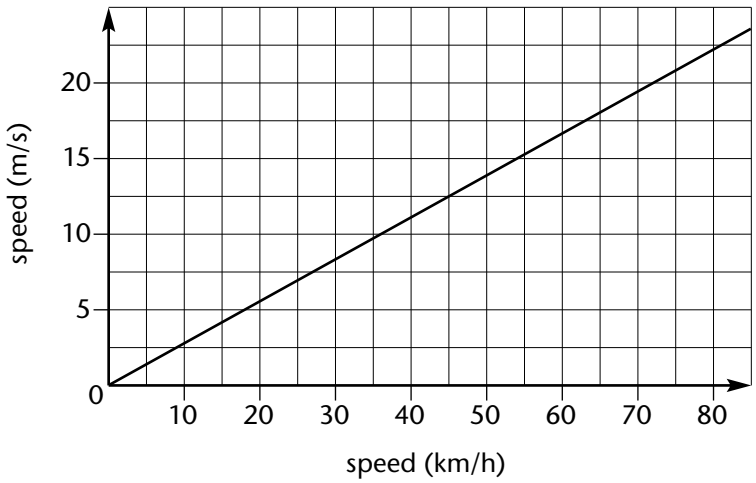
A common acceleration unit is metres per second per second (m/s/s or m/s^2).

In addition to showing speed and time, speed–time graphs can also indicate constant speed, acceleration and deceleration.

Friction is a force. A friction force always opposes the motion of bodies it acts upon.

All objects have inertia. The Law of Inertia, Newton's first law, states that objects remain stationary or continue to travel at a constant velocity – unless acted on by an unbalanced force.

Speed conversion graph



1 State Newton's first Law of Motion, the Law of Inertia.

_____

2 Why do we need to wear seat belts in a moving car?

_____

_____

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Speed and things

6b

3 A passenger was standing at the front of the bus waiting to get out. The bus driver stopped suddenly, throwing the passenger forward. Explain what happened.

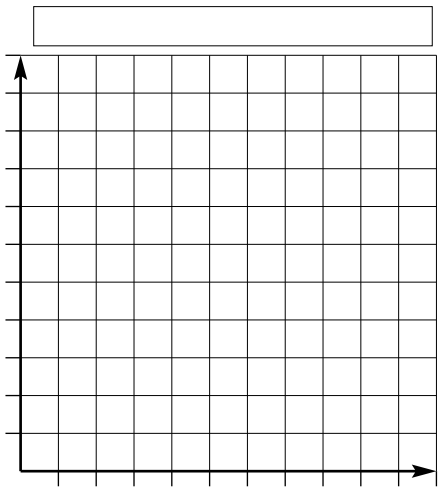
_____

_____

_____

4 Use the information from the table below to draw a speed–time graph. Add a title to your graph. (Hint: Remember, time goes on the horizontal axis.)

Time (s)	Speed (m/s)
1	10
2	10
3	10
4	7
5	5
6	3
7	4
8	6
9	8
10	8



5 Between what times was the car:

a moving at a constant speed? _____

b accelerating? _____

c decelerating? _____

6 A moving car accelerated uniformly for 12 seconds. Before it started to accelerate, its velocity was 6 m/s. When it finished accelerating, its velocity was 24 m/s. What was the car's acceleration?

_____

_____

7 A car was moving at 25 m/s and then quickly decelerated to 0 m/s in 5 seconds because a dog ran across the road. What was the car's:

a initial velocity? _____

b final velocity? _____

c acceleration? _____

d Is the answer to Question 7c positive or negative? Why?

_____

_____

8 **a** On the next page there are three velocity–time graphs. Indicate the graph (use the graph number) that represents a car:

– moving with constant velocity _____ – accelerating _____ – decelerating _____

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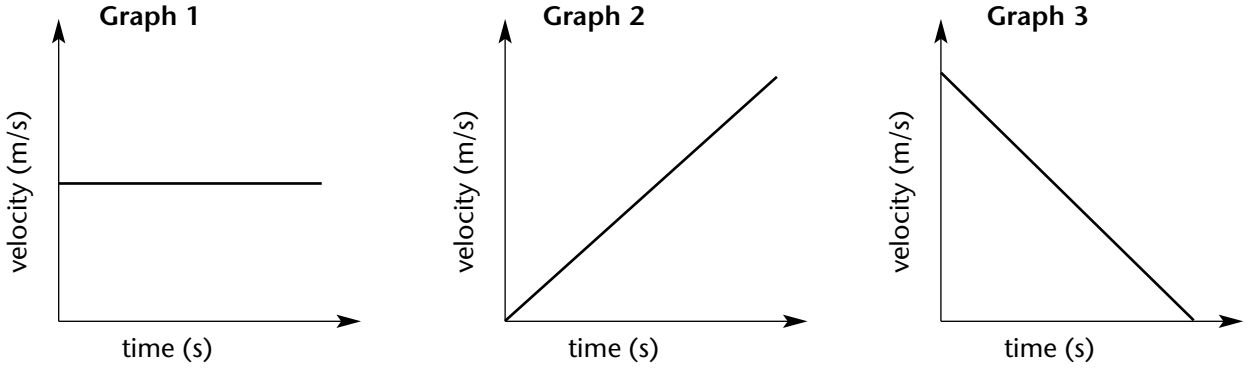
Speed and things

6c



b Give a reason for each choice you made in part a. Identify the feature of each graph you used to make your selection.

9 A car was travelling at 18 m/s and then accelerated at 4 m/s^2 until its speed was 30 m/s. How long did this take? Show your working.

10 What is the difference between speed and velocity?

11 Given distance travelled and time taken to travel the distance, what is the equation for calculating speed?

12 State two uses for each of the following graphs:

a speed–time graphs

b distance–time graphs.

13 A car was driven to Brisbane, a distance of 1200 km, in 20 h. What was the average speed of the trip?

14 Express each of the following speeds in the units indicated in brackets:

a 60 km/h (m/s) _____

b 100 km/h (m/s) _____

c 200 m/s (km/h) _____

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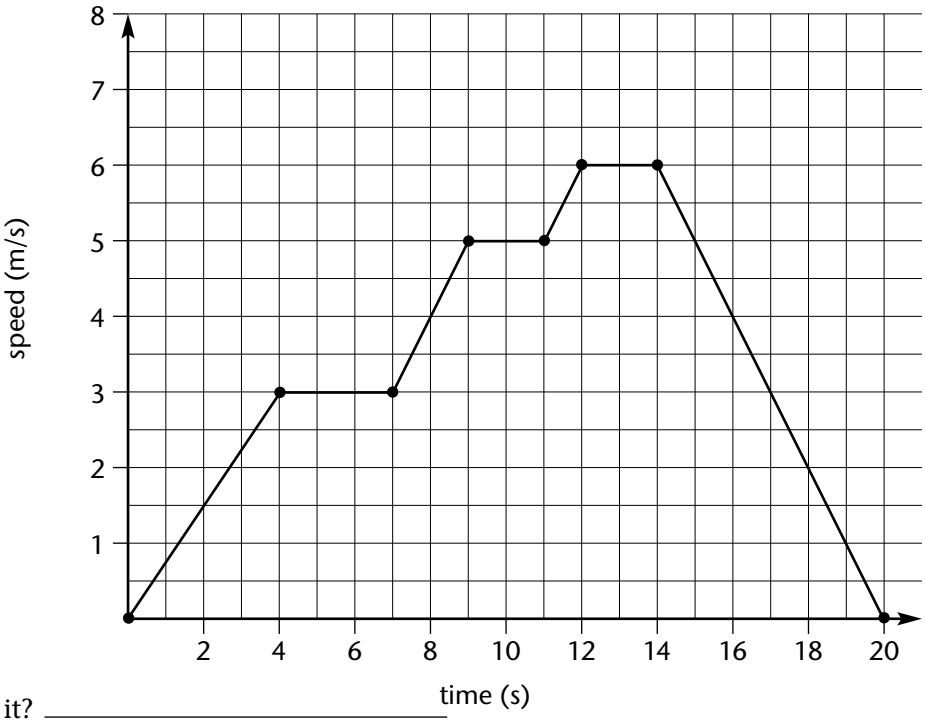
Speed and things

6d

15 Complete the table below, which relates distances, times and average speeds. Show your working out.

Distance travelled	Time taken	Average speed	Calculation/working out
20 km	2 h		
1000 m	20 s		
50 km	5 h		
5000 m	5 min		
	2 h	60 km/h	
	30 s	200 m/s	
30 m		10 m/s	
600 km		60 km/h	

16 This graph represents the motion of a bike over a 20-second time period.



a What type of graph is it? _____

b During which time intervals was the bike travelling at constant speed?

_____

c What was the velocity of the bike at 12 seconds?

_____

d How long did it take the bike to reach a velocity of 5 m/s?

_____

e Fill in the table at right, showing the data from the marked points on the graph. Don't forget to label the columns.

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Car control

7a

In this activity we are using graphs to analyse Chris's off-road driving test. First we are developing a distance–time graph, then a speed–time graph, and finally an acceleration–time graph.

As part of preparing for her licence, Chris had the opportunity of developing her car control through a series of off-road driving tests devised by her instructor. Her instructor placed a line of witches hats 100 metres apart along the edge of the driving path. Smaller markers were positioned midway between successive witches hats. Chris's instructions were as follows.

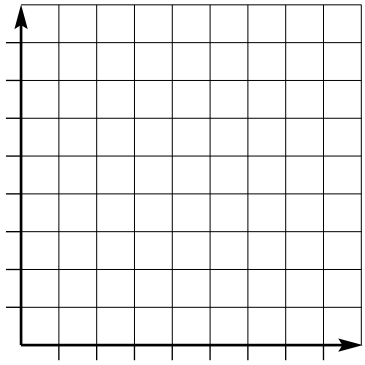
1. Start at zero time at the first hat and move with smooth acceleration to the first small marker in 10 seconds.
2. Drive on at constant speed past the second hat to the second small marker in the next 10 seconds.
3. Brake smoothly to stop at the 200 m mark (3rd witches hat) in 10 seconds.
4. Wait for 10 seconds.
5. Drive smoothly to the next hat in 15 seconds.
6. Brake over 10 seconds to come to rest in 50 metres.
7. Discuss your skill development in the stationary car for 10 seconds.

Developing a distance–time graph

1 **a** Enter the time and distance data from the instructions for the test in the table below.
b Label the axes of the graph grid provided and mark points on the graph to correspond with the data in your table. (Place time on the horizontal axis.)
c Notice that there are seven sections to the graph. Label these sections from A through to G.

Time (s)	Distance (m)

Distance–Time Graph



2 Look at your graph and answer these questions.

a What is the car doing in the graph segments where the graph is horizontal?

_____

b What is happening with the car during the graph segments where the graph is turned upwards?

_____

c When the curve is steepest, what is happening with the car?

_____

d What is happening with the car during the graph segments where the graph slopes a little?

_____

e The slope of the curve tells you at what speed the car is travelling. Speed is the change in distance in a certain time. In which section (A to G) is the speed:

– zero? _____ – lowest? _____ – highest? _____ – uniform but not zero? _____

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Car control

7b

Developing a speed–time graph

Speed is the change in distance over a certain time.

3 **a** What is the formula for finding speed? _____

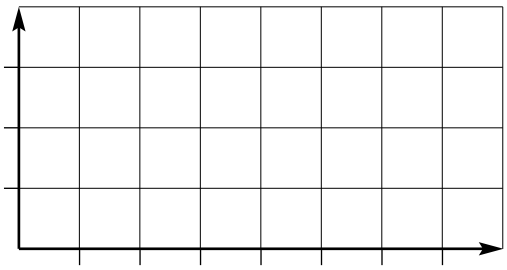
The table below provides space for entering data and calculating the speeds in the different sections of the test drive. Values have been shown in the first line of the table, and the average speed calculated for the starting segment of the test. Below the table there is an explanation of the entries made for this initial line.

Section	First time t_1 (s)	Second time t_2 (s)	Time $t_2 - t_1$ t (s)	First distance d_1 (m)	Second distance d_2 (m)	Distance $d_2 - d_1$ d (m)	Distance / time calculation	Average speed (m/s)	Final speed (m/s)
A	0	10	10	0	50	50	50/10	5	10
B	10								
C	20								
D	30								
E	40								
F	55								
G	65								

Section A: The time interval starts at time zero and extends for 10 seconds. The time length for Section A is then $t_2 - t_1$ = 10 – 0 = 10 s. During that time the car moved from 0 m to 50 m, i.e. it moved 50 m – 0 m = 50 m ($d_2 - d_1 = d$).
The average speed is obtained by dividing the distance travelled by the time taken ($v = d/t$), i.e. 50 m/10 s to give an average speed of 5 m/s. Note that at 10 s the final speed is 10 m/s. The 5 m/s speed is the average of the initial speed 0 m/s and the final speed 10 m/s for the section. Refer to Chris's instructions for the final speed.

b Complete the table above. Enter the known times and distances and calculate the speeds for the different sections of the test.

c Take the data on final speeds from the above table and mark the final speed, second time coordinates on the graph below. Make sure you label the axes with titles, units and numbers as appropriate.



4 Look at your speed–time graph and answer the following questions.

a When the curve in the graph is horizontal, what is the car doing?

_____

_____

b When the curve is sloping upwards, what is the car doing?

_____

c When the curve is sloping up most steeply, what is the car doing?

_____

d When the curve is sloping up but not most steeply, what is the car doing?

_____

e When the curve is sloping downwards, what is the car doing?

_____

f When the curve is sloping downwards most steeply, what is the car doing?

_____

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Car control

g The curve can be divided into seven sections. Each section is a straight line. Start from the left and label the sections A, B, C, D, E, F, G. Note that section A goes from 0 s to 10 s (at 10 s the car has reached a speed of 10 m/s). Section B covers the car's speed from 10 s to 20 s.

h The slopes of the curve show what accelerations the car experiences. Acceleration is the rate of change in speed in a certain time.
In which section (A to G) is the acceleration (i.e. positive acceleration):

– zero? _____ – lowest? _____ – highest? _____

In which section is the deceleration (i.e. negative acceleration):

– smallest? _____ – greatest? _____

Developing an acceleration–time graph

Acceleration is the change in speed over a certain time.

5 **a** What is the formula for finding accleration? _____

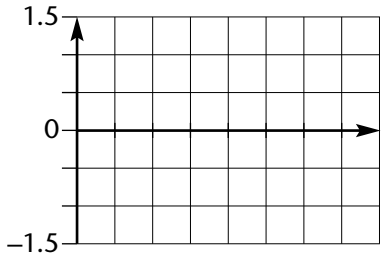
The table below provides space for entering data and calculating the accelerations in the different sections of the test drive. Below the table there is an explanation of the entries made for A, the first section of the test.

Section	Initial time t_1(s)	Final time t_2(s)	Time $t_2 - t_1$ t (s)	Initial speed v_1 (m/s)	Final speed v_2 (m/s)	Speed $v_2 - v_1 = v$ (m/s)	Speed/ Time calculation	Average acceleration (m/s^2)
A	0	10	10	0	10	10	10/10	1
B	10							
C	20							
D	30							
E	40							
F	55							
G	65							

Section A: The time interval starts at time zero and extends for 10 seconds. During that time the car's speed went from 0 m/s to 10 m/s. The acceleration during this period is given by the change in speed divided by the time for the speed change: 10 m/s/10 s = 1 m/s^2. For the first 10 seconds of the test the average acceleration was 1 m/s^2.

b Complete the table above about acceleration. Enter the known times and speeds and calculate the accelerations for the different sections of the test.

c Take the data on acceleration from the above table and mark the acceleration–time coordinates on the graph at right. Label the graph axes with a title, units and numbers.



6 **a** When sections of the curve are in the positive, what is the car doing?

_____

b When sections of the curve are on the x-axis, what is the car doing?

_____

c When sections of the curve are in the negative, what is the car doing?

_____

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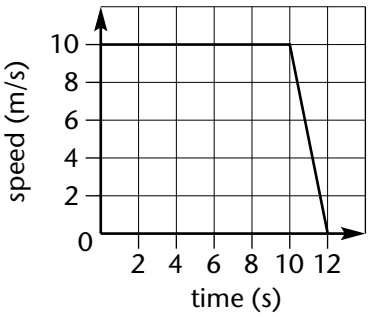
Graph stories

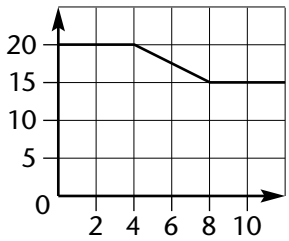
8a

1 Match the graph with its corresponding text by drawing lines between them.

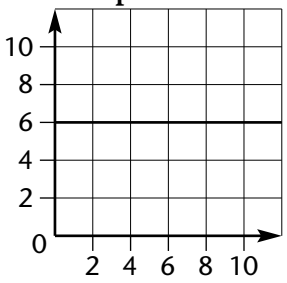
Graph a



Graph b



Graph c

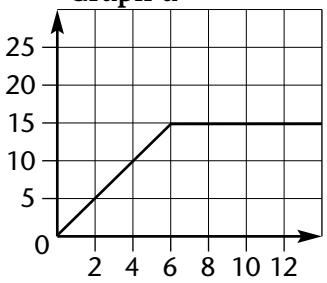


Jane started her car. She accelerated steadily until she reached a speed of fifteen metres per second. This took six seconds.

Chris was in a bike race. The pedal was not right and it took two seconds to get started. Then he accelerated to ten metres per second in four seconds.

Kim travelled at ten metres per second on her skateboard for ten seconds. She rolled onto grass and stopped after two seconds.

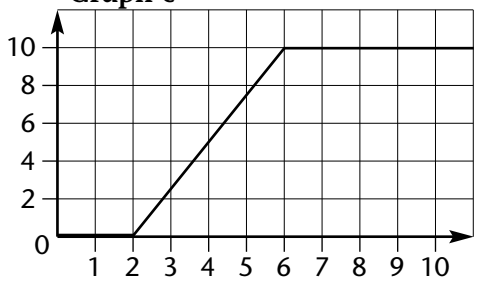
Graph d



Mia ran at a steady speed of six metres per second for twelve minutes.

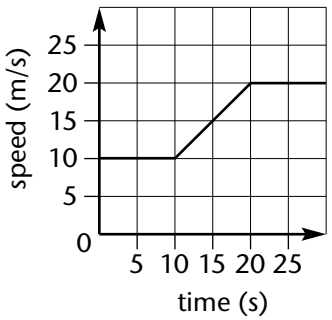
Sam was driving at a speed of twenty metres per second and noticed a speed sign. He decelerated in four seconds to a speed of fifteen metres per second.

Graph e

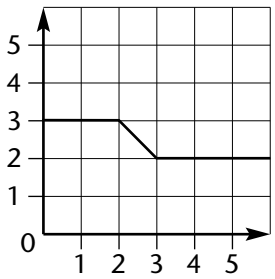


2 Write a brief story to accompany each of the following speed–time graphs.

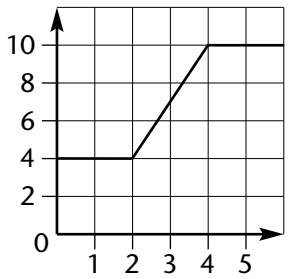
a Mick's rollerblade journey



b Alex's walk



c Sam's run



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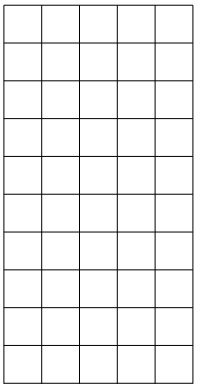
Graph stories

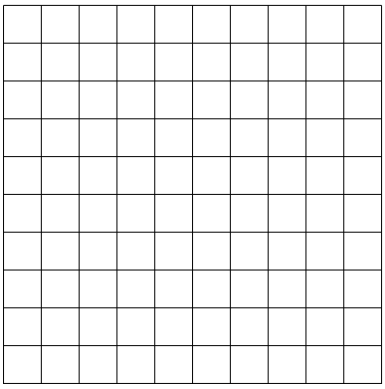
8b

3 Read each story and then draw a labelled graph about the speed.

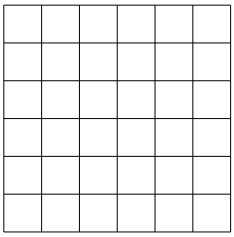
a Nik rode on her scooter at a constant speed of eight metres per second for thirty seconds. In order to get a drink of water, she slowed to a stop taking thirty seconds to do so. After one minute she pushed off on her scooter again and accelerated to a speed of ten metres per second in one minute. She kept cruising at ten metres per second for two minutes.



b Jo jogged for two minutes at a speed of five metres per second. Then on a downhill slope he took one minute to increase his speed to eight metres per second. He stayed at this speed for two minutes. When he reached a steep hill, he took one minute to slow his speed to three metres per second. He maintained that speed for two minutes and then gradually slowed to a stop during the next two minutes.

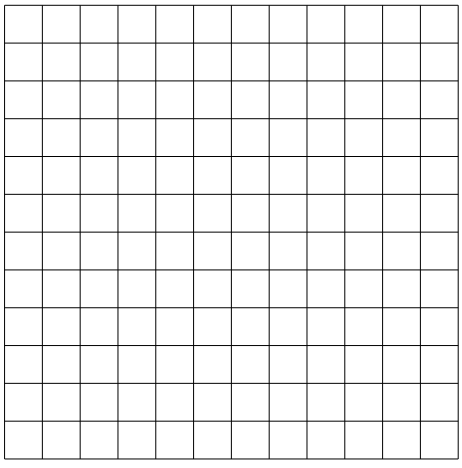


c Maxi shot an arrow into the air. When it left the bow, it had a vertical speed of thirty metres per second. It rose for three seconds until it reached its highest point and stopped. Then it began falling with an increasing downward speed. After three seconds, it hit the ground at a speed of thirty metres per second.



4 Graph the information in the table and then write a paragraph about Lee's car trip.

Time (s)	Speed (km/h)
0	50
10	50
20	40
30	20
40	0
50	0
60	0
70	10
80	20
90	30
100	30



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Speed and velocity – what's the difference? 9a

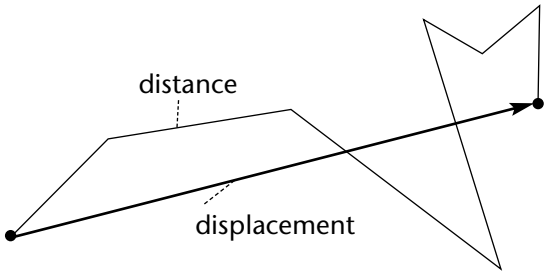
1 Read through the following paragraphs. Cross out the incorrect terms in brackets.

Speed and velocity (mean / means) almost the same (thing / think) but (their / there) is a slight distinction between the (too / two) terms. For example, if we write that a car (travelled / travels) at a constant (rate / rates) of 60 (km/h / kph), we are specifying (its / it's) speed. But if we (say / says) that the car (are travelling / is travelling) at 60 (kmh / km/h) to the (south / South), we (are talking / is talking) about velocity. Therefore, when we (specify / spicify) the speed (and / or) the direction, we (are / is) talking about (speed / velocity). Both speed (and / or) velocity (are / is) represented by v.

speed: v = 60 km/h velocity: v = 60 km/h south

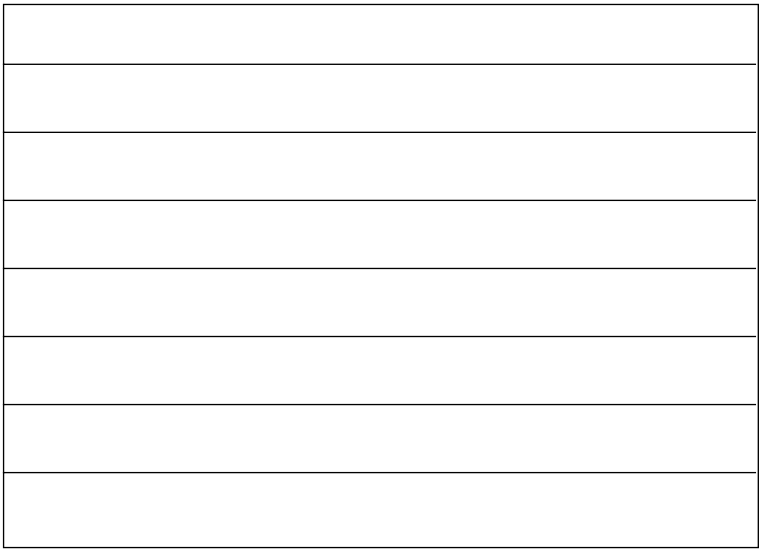
A similar distinction (exist / exists) between distance and displacement. Distance (are / is) about how much ground (has been covered / has covered) during a trip, while displacement (staits /states) how far out of place an object (is / was). In other words, distance (has / is) the travelled length of the trip and displacement (are / is) about the object's change in position. Displacement (tells / tels) how far the object (are / is) away from its original position and in which direction the new place (is / was). Displacement (is / was) the shortest distance between start (but / and) finish. The symbol for distance (has / is) d and the symbol for displacement is s.

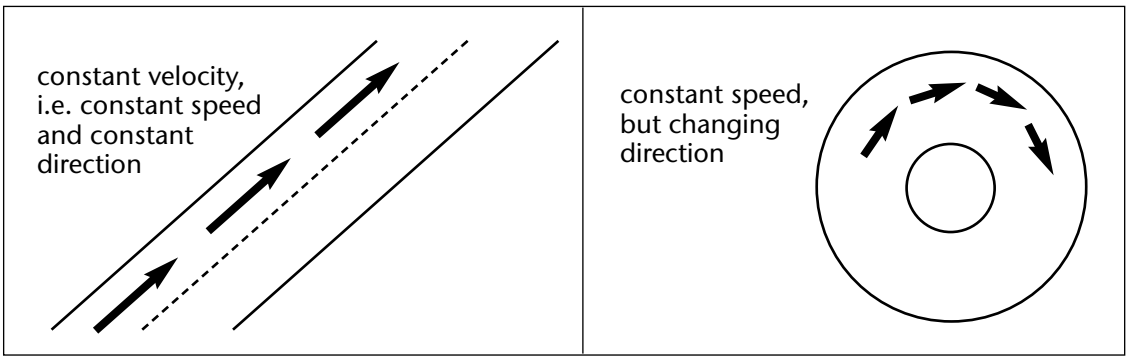
Consequently, speed (calculates / is calculated) by dividing the distance the object has travelled by the time taken to travel this distance. Velocity (finds / is found) by dividing the object's displacement by the time taken to make the displacement.



2 Cut out the sentence fragments on page 25, sort them into their correct order and then glue them into the spaces below. There are two paragraphs, a and b.

Paragraph a





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Speed and velocity – what's the difference? 9b

Paragraph b

3 Draw lines to join each concept to its correct symbol, unit and formula.

Concept	Symbol	Unit	Formula
displacement	d	m	d/t
distance	s	m west	s/t
speed	t	s	s/v
time	v	m/s	vt
velocity	v	m/s east	vt

4 Define each of the following:

a distance _____

b displacement _____

c speed _____

d velocity _____

e average speed _____

f instantaneous speed _____

5 Draw ticks or crosses in the cells to show whether the features of size and direction are associated with each concept.

Concept	Has size?	Has direction?
distance		
displacement		
speed		
average speed		
instantaneous speed		
velocity		
average velocity		
instantaneous velocity		

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Speed and velocity – what's the difference? 9c

6 Complete the following table by relating distance, displacement, time, speed and velocity where appropriate.

Distance/ Displacement	Time	Speed	Velocity
60 km	2 h		
1 500 m E	30 s		
120 km S		40 km/h	
400 m		80 m/s	
	5 h		50 km/h N
	40 s		20 m/s W

Sentence fragments to be cut out for Question 2

Paragraph a:

along a curved path like a roundabout. But

Constant speed and constant velocity also are

for example, a car driving along a long straight

its direction is changing continually in the roundabout.

moving at constant speed in the same direction;

not the same. Constant velocity describes an object

path. However, a car can move at a constant speed

this car does not have constant velocity, because

Paragraph b

by the time it took to cover that distance. Instantaneous speed

downhill, your instantaneous speed might well have been 30 km/h.

example, imagine you were cycling to a friend's house. It took you

is the speed at which an object is travelling at a particular time. For

km/h. But when you stopped at the

lights, your instantaneous speed was 0 km/h, and when you raced

one hour to travel 10 kilometres, so your average speed is 10

Sometimes average speed and instantaneous speed are confused. Average

speed is found by dividing the travelled distance

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Travelling fast

10a

Speed up your calculations by using a calculator.

1 Speed is how fast you are going. It is defined as how far you go in a unit of time. **Speed** is calculated by dividing the **distance** travelled by the **time** taken to travel the distance.
Use the bold words above to write a formula for calculating speed.

_____

2 Distance (i.e. how far) is often measured in **metres** or **kilometres**, while time (i.e. how long) is often measured in **seconds** or **hours**.
In the table below write names and abbreviations for the bold units for distance and time.

Distance units		**Time units**	
Name	**Abbreviation**	**Name**	**Abbreviation**

3 Suggest useful units for measuring the speeds involved in the situations listed in the table below.

Situation	**Possible unit**	**Abbreviation**
A car driving on a highway		
A rocket flying through space		
A person running a short race		
A snail crawling along the ground		

4 To compare two speeds, the speeds must have the same unit, such as km/h or m/s.

a A speed of 1 m/s is the same speed as a speed of:

_____ km/h.

b Using the information in the box at right as a model, write an expression for 1 km/h in terms of m/s.

Converting 1 m/s into km/h
Recall: 1 m = 1/1000 km = 0.001 km
1 h = 60 . 60 s = 3600 s; 1 s = 1/3600 h

Substitute 1 m with 1/1000 km and 1 s with 1/3600 h.

1 m/s = 1 . 1/1000 . 3600 m/h = 1 . 3600/1000 km/h
= 3.6 km/h

1 km/h = _____

_____

A speed of 1 km/h equals _____ m/s.

c Express each activity (distance–time relationship) below as a speed, in both m/s and km/h. Show your working out.

Activity	**Calculating speed in m/s**	**Calculating speed in km/h**
Walking 10 km in 2 h		
Jogging 400 m in 60 s		
Running 100 m in 20 s		
Swimming 50 m in 60 s		
Cycling 50 km in 2 h		

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Travelling fast

10b

5 Sometimes your speed is known. Knowing speed lets you calculate the time needed to cover a certain distance.

a Write the formula for speed. _____

b Rewrite the formula so that time becomes the subject of the formula.

_____

c Calculate the time needed to cover each distance listed in the table. The speeds to be used for each distance are indicated in the speed column.

Speed	Distance to be covered	Calculation of time needed
School speed limit: 40 km/h	0.2 km past a school (200 m)	
Town speed limit: 60 km/h	2 km along the main road	
Sydney Harbour Bridge speed limit: 70 km/h	500 m across the bridge	
Freeway speed limit: 110 km/h	300 km from Sydney to Canberra	

6 Some careful drivers say a 2 s time gap should be left between moving cars. Obviously this time gap equates to bigger distances as cars travel at higher speeds. When cars are travelling at 30 km/h, what distance is represented by a 2 second gap? To express the 2 second gap in terms of metres, the speed of 30 km/h can be converted to m/s and then used to find the distance the car would travel in 2 s. Recall that 1 km/h = 1/3.6 m/s. Hence 30 km/h = 30 . 1/3.6 m/s = 8.33 m/s. A 2 s time gap corresponds with a distance of 8.33 . 2 m = 16.66 m. This distance divided by the length of a car (i.e. 16.66/4 = 4.17) tells you that you should be more than 4 car lengths behind the car in front of you.

a Write down the formula for speed.

_____

b Rearrange the formula and make distance the subject.

_____

c Now complete the table, showing your calculations as well as the answers. The table shows the time gap measured in seconds, and the distance gap in metres and in car lengths (assuming that a car is 4 m long like a medium-sized car).

Speed (km/h)	Speed (m/s)	Time gap (s)	Distance gap (m)	Distance (4m car lengths)
30	30/3.6 = 8.33	2	8.33 . 2 = 16.66	16.66 / 4 = 4.17
50				
80				
110				

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Stopping really fast

11a

1 How well a car stops depends on a whole lot of things. Draw ticks in the table to show whether the listed feature helps a car to stop more quickly or whether the feature slows the stopping.

Feature	Assists faster stopping	Hinders faster stopping
Worn tyres		
Wide tyres		
Five people in the car		
Only the driver in the car		
Lots of luggage		
Very loud music		
Slow music		
Small car		
Heavy car		
Disc brakes		
Drum brakes		
Combined disc/drum brakes		
Big brake pads		
ABS brakes		
Tired driver		
Alert driver		
Distracted driver		
Low speed		
High speed		
Racing car driver		
Sunday driver		
Car with caravan		
Smooth bitumen road		
Dirt road		
Grassy road		
Wet road		
Road during a light shower		
Dry road		

How quickly a car stops depends on the reaction time of the driver (the time before the driver acts to slow the car) and on the braking time (when the driver is actually pushing down on the brake pedal).

2 Write a formula for calculating the stopping distance.

_____

Assume that the reaction time for a driver is one second. This reaction time would change very little for a particular driver and it doesn't depend on the driver's speed. However, what does change is how much ground is covered in that one second when the car is travelling at different speeds. The distance covered by a car in one second at various speeds becomes very important when considering stopping times.

3 Write the formula for speed. _____

4 Rearrange the speed formula and make distance its subject.

_____

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Stopping really fast

11b

Remember when carrying out calculations that all units in your calculations must be matching units. Sometimes it is necessary to convert one unit to another to maintain consistency of units in calculations. Remember: To convert m/s to km/h, multiply the number of metres per second by 3.6 (1 m/s = 3.6 km/h). To change km/h to m/s, divide the kilometres per hour by 3.6 (1 km/h = 1/3.6 m/s or 0.28 m/s).

5 Calculate the reaction distances for a driver with a reaction time of 1 second for the speeds shown in the table below. Show your working out.

Speed (km/h)	Speed (m/s)	Time (s)	Reaction distance (m)
20		1	5.56
40		1	
50		1	
60		1	
80		1	
110		1	
200		1	

A braking test was carried out at a driver training school.

A driver drove a car to a certain speed and then pressed the brake pedal to stop the car as quickly as possible (without locking the wheels). At a low speed of 20 km/h, the car stopped within 3 metres. At 40 km/h, the braking distance was 10 metres. At 50 km/h, the braking distance was 15 m and at 60 km/h it was 22 m. At 80 km/h the car needed 38 m to come to a stop and at 100 km/h the braking distance had gone up to 60 m. At the NSW highway top speed of 110 km/h, the car took 72 m to stop.

6 Summarise the results of the braking test in the table at right.

7 Construct another table (below) to summarise the information gathered in this activity. The table should consider at least five different speeds and show the reaction distance, the braking distance and the total stopping distance for these five different speeds for a driver with reaction time of 1 second.

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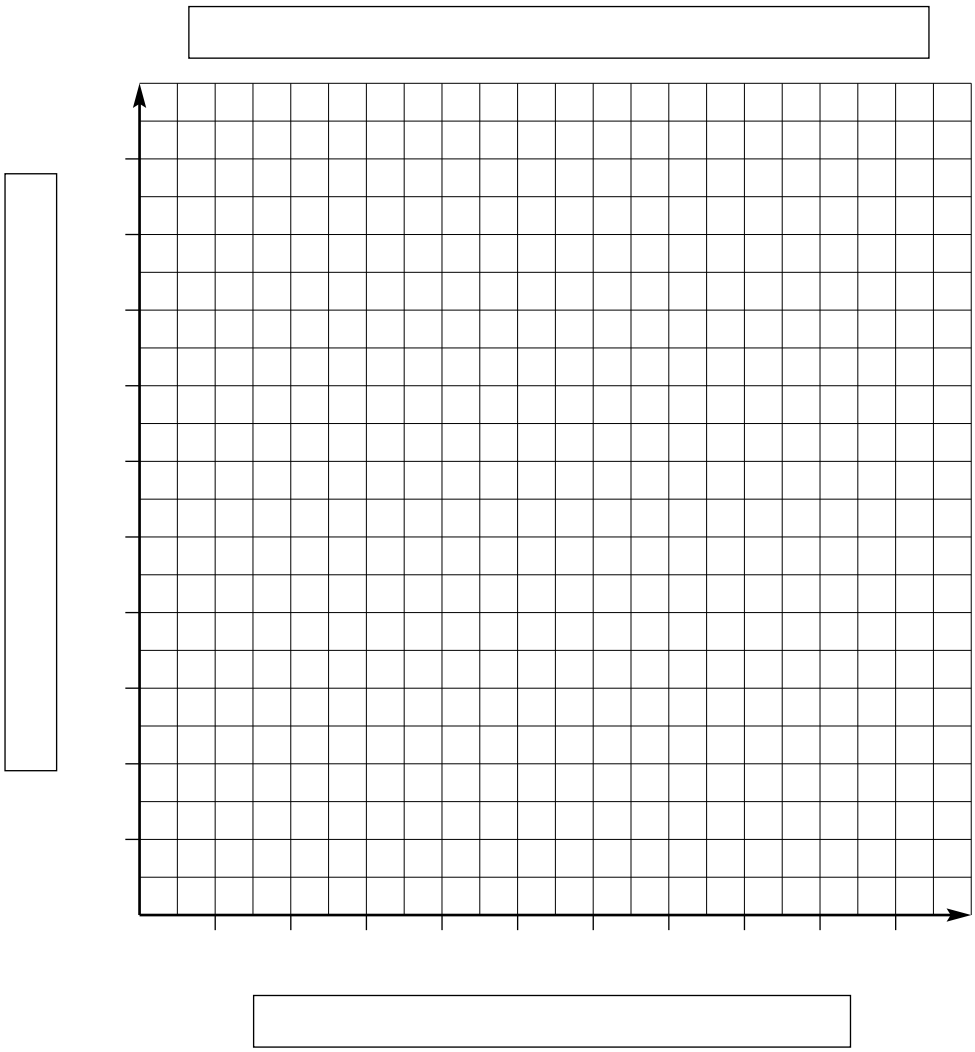
Stopping really fast

11c

8 Use your completed table from question 7 to graph the reaction distance, braking distance and stopping distance for each speed. Plot speed on the horizontal axis and distance on the vertical axis. To highlight the different distances, use a red line for stopping distance, a blue or dotted line for reaction distance and a black or dashed line for braking distance. Label the axes and the curves. Give a title to your graph.



9 In considering reaction time and braking distance, everyone can formulate some personal driving advice. Complete the sentence below.

The _____ I drive the _____ distance I need to keep from the car in front and the _____ alert I must be.

10 Sometimes road signs are designed to alert drivers to changes in road conditions or possible dangers. Use the back of this sheet to draw the road signs that would be used for each of the following situations.

a approaching a school

b pedestrian crossing ahead

c approaching a railway crossing

d stock crossing ahead

e children's playground nearby

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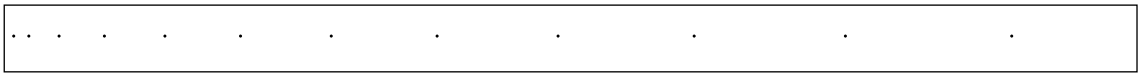
Ticker tapes and oil drops

12a

A thin paper tape passing through a buzzer was attached to a mechanical mouse. The idea was to use the buzzer and its imprint on the tape to make a record of the mouse's speed over a period of time. The buzzer makes an impact on the tape once every tenth of a second; so ten impacts on the tape mark the passing of one second. Below is the beginning of one of these tapes.



1 Describe the trend about the distance between dots as the mouse moved along.

_____

2 Number the gaps between two adjacent dots (staring at the left of the tape with the smallest gap being numbered '1').

3 Measure the distance between every two dots and record the gap distances in the table below.

Gap number	1	2	3	4	5	6	7	8	9	10	11
Gap distance (mm)	2										
Gap time (s)	0.1										
Average speed (mm/s)	20										

4 Complete the row about gap times.

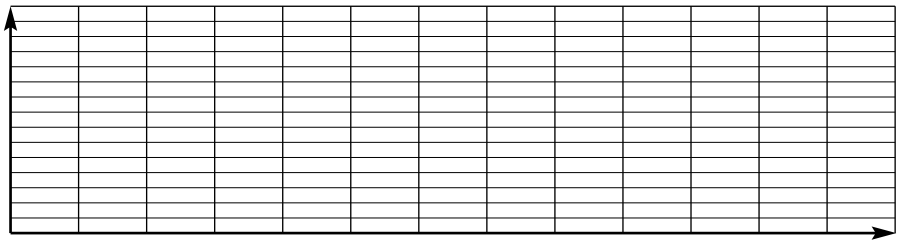
5 State the formula for calculating speed. _____

6 Calculate the average speed for every gap and write it in the above table.

7 Describe the trend about the speed as the mouse moves along.

_____

8 **a** Cut across the tape at every dot. Glue the pieces in order from smallest to largest onto the graph below. Label the axes of the graph appropriately, and give it a title.



b Connect the dots at the top of the tape pieces and draw a smooth line through them.

c State what the smooth line represents.

_____

d Calculate the overall average speed from your graph.

_____

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Ticker tapes and oil drops

12b

9 Acceleration can be found by subtracting the initial speed from the final speed, and dividing the answer by the time between the speed readings. Write the formula for calculating acceleration.

---

10 **a** Calculate the acceleration at three different places in your graph on the previous page. Use the table below as guide.

Initial speed (mm/s)	Final speed (mm/s)	Speed difference (mm/s)	Time for speed difference (s)	Acceleration (mm/s/s)

b Describe the trend about the acceleration as the mouse moves along.

---

11 A car travelling along a road has been dropping oil, one drop of oil every second. Part of the oil trace is shown on the following scaled diagram.

(Key: 1 mm on the drawing = 1 m on the road.)

Start Finish

. . . . . .

a Indicate between successive drops whether the car was accelerating (ac), decelerating (de) or travelling at a constant speed (co). Mark these symbols on the diagram above.

b Calculate the car's speed at six different places along the oil trail. Number your chosen places 1, 2, etc. Use the table to record distances, calculations and speeds.

Interval	Distance (drawing)	Distance on road	Distance/time	Average speed
1	3 mm	3 m	3/1 ms^{-1}	3 m/s
2				
3				
4				
5				
6				

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Cars and gears

13a

In a driving performance test, a car was put through its paces in order to find out the top speeds for the various gears. In first gear, it took three seconds to reach a speed of fifty kilometres per hour from standstill. Following a quick change into second gear, in another four seconds a speed of eighty kilometres per hour was reached. Third gear allowed a top speed of one hundred and ten kilometres per hour in a further six seconds. It took eleven seconds in fourth gear to reach one hundred and forty kilometres per hour. To reach the top speed of one hundred and sixty kilometres per hour required thirteen seconds in fifth gear. Finally, it took thirteen seconds of hard braking to bring the car to a complete stop.

1 What is the top speed of the car? _____

2 How long did it take the car to reach its top speed? _____

3 Which gear has shown the biggest speed change? _____

4 Which gear enables the fastest changes in speed? (Gives the greatest acceleration?) _____

5 Make a table of the results using these headings: Gear, Initial speed, Final speed, Speed change, Time taken, Total time. (Don't forget to include the units in the headings!)

Gear					

Acceleration is defined as the rate of change in velocity over time. (Velocity has the same magnitude as speed, but has a direction as well.) In our case, the speeds have been measured in kilometres per hour and the times in seconds. Therefore the unit for acceleration here is kilometres per hour per second, or km/h/s. We can find the change in speed by subtracting the initial (i.e. first) speed from the final (i.e. second) speed.

6 **a** Write a formula to find the change in speed.

_____

b Scientists usually write the letter **a** for acceleration, **t** for time, and **v** for speed. Initial speed is $\mathbf{v_1}$ and final speed $\mathbf{v_2}$. Rewrite your formula for the change in speed using abbreviations.

_____

7 State a unit for speed _____ and a unit for acceleration _____.

8 **a** To calculate acceleration, the change of speed is divided by the time it took to change that speed. Write a formula to calculate acceleration.

_____

b Rewrite your formula for acceleration using abbreviations. _____

9 Calculate the acceleration for each gear. (Don't forget to show your working out!)

Gear	Speed change	Speed change/Time taken	Acceleration
First			
Second			
Third			
Fourth			
Fifth			

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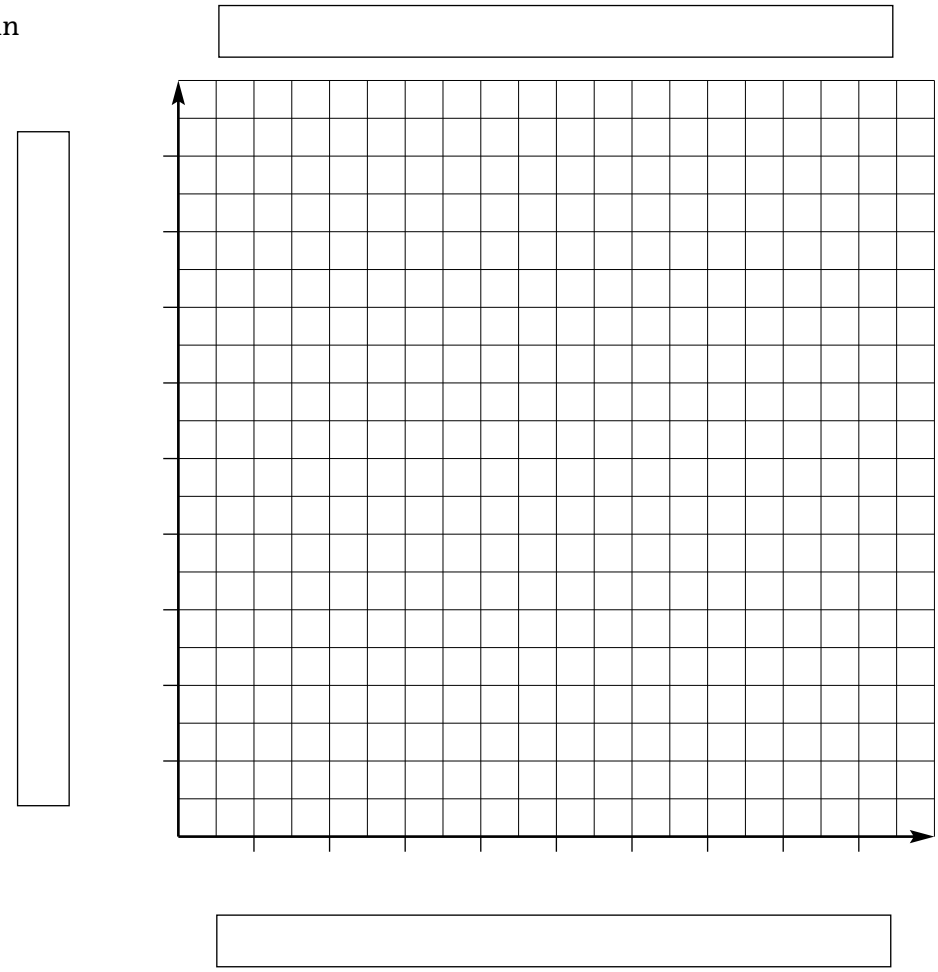
Cars and gears

13b

10 Record the speed results in a speed–time graph. (Hint: Time goes on the horizontal axis.)



11 **a** State how you can tell from the graph which gear produces the greatest acceleration.

b State how you can tell from the graph which gear produces the smallest acceleration.

c Explain how you could calculate the acceleration produced by a particular gear from the graph.

12 **a** Draw a line on the graph to represent the average acceleration of the car as it travelled from standstill to top speed.

b Calculate the average acceleration of the car. _____

13 A positive acceleration occurs when the car is becoming faster – the speed increases as time passes, so the final speed is higher than the initial speed. A negative acceleration (or deceleration) happens when the car is slowing down – the speed decreases as time increases, so the final speed is lower than the initial speed.

a State how you could identify a positive acceleration on a speed–time graph.

b State how you could identify a negative acceleration on a speed–time graph.

c Describe what no acceleration (i.e. a constant speed) looks like on a speed–time graph.

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Baby on board

14a

When a baby travels in a car, the baby must be in a specially designed baby or child restraint. That restraint helps to protect the baby by keeping him or her in place as the car changes speed and/or direction. It also provides maximum protection if there is an accident. Imagine that a car is completely full, with all seats occupied. Someone does an illegal, dangerous and stupid thing, taking the 6 kg baby to hold on their lap. There is an accident, with the car hitting a brick wall at a speed of 20 km/h.

We will calculate what force is needed to keep the baby on the holder's lap when a car comes to a sudden stop. Newton's second law of motion can be used here. Newton found that multiplying the mass of an object (*m*, measured in kg) and its acceleration (*a*, measured in m/s/s) gives the force needed to bring the object to rest (*F*, measured in N).

1 Write Newton's force equation. _____

Assume that a car takes a quarter of a second to come to a complete stop after hitting a wall. (As soon as the car hits the wall, the bumper bar and the metal begin to crumple and it takes some time for the car to be at a total standstill.)

2 Calculate the change in speed of the car. _____

Note that the negative change in speed shows that the speed is decreasing.

3 Convert the speed change in km/h to a speed change with the unit m/s. (Hint: Divide the km/h speed by 3.6 to express it in m/s.)

_____

4 Write the formula for calculating acceleration.
(Hint: Acceleration is defined as the change of velocity in a unit time.) _____

5 Calculate the acceleration of the car (and anybody wearing a seat belt in the car!). The unit of acceleration will be m/s/s (or m/s^2, as it is most often written).

_____

_____

Note that the acceleration is negative. Because it is a negative acceleration, it is called a deceleration. The negative sign indicates that acceleration slows the car. The negative sign is often dropped in calculations and remembered as the direction of the acceleration.

6 Substitute the value for the acceleration in Newton's force equation to find the force that is needed to keep hold of the 6 kg baby.

_____

_____

7 If this force was a weight and pulled downwards (instead of horizontally, as it would if the baby were on a person's lap), we can calculate an equivalent mass that would experience the same force. In other words, we can equate the deceleration force on the baby with the mass of an imaginary baby being held up against the force of gravity.

a Write Newton's force formula again. _____

b Rearrange the formula so that mass is its subject. _____

c The size of the force is known from Question 6. Acceleration due to gravity is 10 m/s^2.
Substitute these values into the mass formula and calculate the mass of the imaginary baby being held up against the force of gravity.

_____

We have just calculated how big the deceleration force on a 6 kg baby appears to be when someone is trying to keep the baby on a lap during a 20 km/h crash. In other words, the baby seems to have a mass of the answer in 7c, when his or her actual mass is only 6 kg.

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Baby on board

14b

8 Use your new understanding from this activity to find how much mass a 6 kg baby would appear to have if a car travelling at 50 km/h crashed and was brought to a stop within a quarter of a second. Show your working.

9 Find how much mass the baby would appear to have if the car was travelling at 80 km/h, crashed and was bought to a stop in a quarter of a second. Show your working out.

10 A different baby has a mass of 10 kg, and was in a car travelling at 110 km/h. The car crashed, coming to a standstill in 0.25 s. What was the deceleration force on the baby? Show your working out.

11 Two passengers in a car chose not to wear their seatbelts. One passenger has a mass of 50 kg and the other passenger has a mass of 70 kg. The car was travelling at 95 km/h when it crashed, coming to a standstill in 0.5 s. What was the deceleration force on each passenger? Show your working out.

12 Imagine a family car (mass of the car plus occupants is 1500 kg) coming to a quick stop (in 0.5 s) from a speed of 150 km/h (some roads in the Northern Territory have no speed limit). Calculate the deceleration force for the whole car. Show your working out.

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Mass and weight – what is which? 15

Mass is defined as the amount of substance in an object. Its base unit is the kilogram (kg).

Weight is a force. It is a force that arises because of gravity. The larger an object, the stronger the gravity force; so huge objects like stars have strong gravity, which causes things to weigh more there than on smaller objects. Weight is measured in newtons (N).

It is common to say 'My **weight** is 50 kg', but this is scientifically incorrect and probably arises from the confusion between mass and weight that many people have. (A person of mass 50 kg actually has a weight of 500 N.) We should actually say 'My **mass** is 50 kg'.

We are going on an imaginary journey through the solar system. As well as food and radiation suits, we are taking a giant triple beam balance for measuring our mass and a bathroom scale for our weight. On earth, the giant triple beam balance shows your mass as 60 kg and the bathroom scale gives your weight as 60 kg (although this is really 600 N!).

1 Let's fly towards the moon. At a certain point between the earth and the moon, the gravitational forces of the earth and moon cancel each other. At that point:

a How strong is the gravitational force on us? _____ (newtons)

b What is your mass? _____ (kilograms)

c What does the bathroom scale show your weight to be? _____ (kilograms)

d What is your weight in scientific measure? _____ (newtons)

2 Now let's land on the moon and take our measuring devices with us onto the moon's surface. The moon's mass is about six times smaller than the earth's mass. This means that the moon's gravity is one-sixth that of the earth's gravity, i.e. the moon's gravity is six times smaller than the earth's.

a What is your mass on the moon, as measured by the giant triple beam balance? _____

b What does the bathroom scale show your weight to be? _____

c What is your scientific weight? _____

d If you can jump 1 metre high on earth, how high could you jump on the moon? _____

3 We're leaving the moon and travelling on to Jupiter, which has a mass about 300 times bigger than the earth.

a About how strong is Jupiter's gravity compared to earth's gravity? _____

b What is your mass on Jupiter, as measured by the giant triple beam balance? _____

c What does the bathroom scale show your weight to be on Jupiter? _____

d If you could lift 30 kg (30 000 g) on earth, what could you lift on Jupiter? _____

4 Onwards to the sun. The sun's surface temperature is about 6 000°C and we would simply burn into a cinder well before arrival; but let's ignore that small detail. The sun's mass is about 300 000 times bigger than the earth's mass.

a About how strong is the sun's gravity force compared to the earth's? _____

b What is your mass on the sun? _____

c What is your weight on the sun? _____

d If you could throw a ball 30 m (= 30 000 mm) into the air on earth, how high could you throw the same ball on the sun? _____

According to Newton's second law of motion, the force on an object depends on its mass and its acceleration, i.e. $F = m \cdot a$. The mass is your body mass (e.g. 60 kg) and the acceleration is the acceleration due to gravity (10 m/s^2 on earth). Therefore the weight force on earth is $F = m \cdot a = 60\ kg \cdot 10\ m/s^2 = 600$ N. Note that on earth, one Newton equals the gravity force on 100 grams; 10 N attract 1 kg.

On Mars, the acceleration due to gravity is only about 1 m/s^2, as Mars has a fairly small mass.

5 **a** What would your weight force be on Mars? _____

b What does the bathroom scale show your weight to be? _____

6 Uranus has a gravity acceleration of about 150 m/s^2, as its mass is much greater than the earth's.

a Calculate your weight force on Uranus. _____

b State your weight as shown on a bathroom scale. _____

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Newton's laws of motion: history

16

In 1686 Sir Isaac Newton published his three laws of motion – laws that form the basis of 'classical physics'. The movement of our daily lives is well described by these laws. When we play sport, the laws predict the outcome. Rocket scientists use the laws to predict the paths of their rockets and space craft. Many physics experiments in schools investigate them.

Although Newton had formulated the laws of motion possibly as early as 1666, it was only in 1686 that he published his single greatest work, the *Philosophiae Naturalis Principia Mathematica* ('Mathematical Principles of Natural Philosophy', simply known as the 'Principia').

In it, Newton revealed his laws of motion and the law of universal gravitation. The results were applied to orbiting bodies like planets and satellites, projectiles like bullets and golf balls, pendulums in clocks, and free-fall near the Earth.

The first law of motion is about inertia: a body wants to keep doing what it is doing. This means that a moving body wants to continue moving in a straight line, while a stationary body wants to remain stationary.

The second law says that the acceleration of an object depends on the force acting on it and its mass. ($a = F / m$)

The third law states that every action force has an equal but opposite reaction force.

Newton demonstrated that heavenly bodies mutually attract one another. Further generalisation led him to the law of universal gravitation: every object has a gravity force that attracts other objects. As the distance between the two objects increases, the strength of the gravity force between them decreases.

The Royal Society was going to publish Newton's book, but withdrew due to lack of funds. The astronomer Edmund Halley (of Halley's comet fame), who was wealthy and thought highly of Newton, eventually paid for its publication in London. The Principia is recognised by some as the greatest scientific book ever written.

1 Why is Isaac Newton often called Sir Isaac Newton?

2 Why might Newton have waited twenty years before publishing his laws?

3 When and where was the book published?

4 Who paid for the publishing costs?

5 What are Newton's laws about?

6 What is the keyword of Newton's first law?

7 According to that law, what does a moving object want to do?

8 What does a stationary object want to do?

9 If an object is moving in a straight line, what does it want to do?

10 If an object is falling towards earth, what does it want to do?

11 What is the formula of Newton's second law?

12 If the force is increased, what happens to the size of the acceleration?

13 If the mass is increased, what happens to the size of the acceleration?

14 What are the forces in Newton's third law?

15 If the action force pushes forward, in which direction does the reaction force push?

16 If the action force measures five newtons, how big is the reaction force?

17 If three action forces, measuring 3 N, 4 N and 5 N, are pushing upwards, how big is the reaction force and in which direction is it acting?

18 Folklore has it that Newton came up with the idea of gravity when an apple fell from a tree and hit him on his head.

a If the earth's gravity attracted the apple, did the apple's gravity also attract the earth?

b If the apple moved towards the earth as it fell, did the earth also fall towards the apple?

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Newton's laws of motion: applications 17a

NEWTON'S FIRST LAW: INERTIA Newton's first law of motion states that an object that is moving will continue to move at the same speed and in the same direction – unless a new force is acting on it. It also states that an object that is not moving will want to stay still – unless a new force is acting on it. This characteristic (not wanting to change) is called **inertia**.

1 In the following situations, state which object is changing its movement or direction (due to a force) and which is not (due to its inertia).

Situation	Changing	Not changing
A bus is taking off from the bus stop; a traveller falls backwards.		
A car is turning right; a passenger is pushed into the left door.		
The car stops suddenly; a bag is thrown against the windscreen.		
Bags fall off the luggage rack into the aisle when a train goes round a bend.		
A plane hits an air pocket; passengers float above their seats.		
A plane takes off; a passenger feels forced back into the seat.		
Your seat belt cuts into you as the car brakes suddenly.		
You sink to the floor as the lift goes up suddenly.		
The car slows down suddenly; the steering wheel pushes against the driver's chest.		
You fly over the handlebars because your bike hits a rock.		

2 Use Newton's law about inertia and your common sense to explain the following.

a People travelling in cars wear seatbelts (it's also the law!).

_____

b Motorbike riders don't wear seatbelts.

_____

c Commuters on city buses don't wear seatbelts, but people on overland coaches do.

_____

d Travellers on planes are asked to put their seatbelts on when the plane hits air turbulence.

_____

e The lift starts to go down quickly and you feel somewhat weightless.

_____

NEWTON'S SECOND LAW: F = m . a Newton's second law is about **acceleration**: acceleration makes an object go faster or slower, or change its direction. Acceleration depends on the force that is acting on the object and the mass of the object. Newton came up with the formula: Acceleration = force/mass or a = F / m.

3 State the units of the three quantities in the formula: a = acceleration: _____;

F = force: _____; m = mass: _____

4 A car's motor produces a force of 7.5 kN. The car has a mass of 1 500 kg. Calculate its acceleration. _____

5 A cyclist, having a mass of 100 kg (including the bike), is able to push on the pedals with a force of 200 N. Find how big the acceleration is. _____

6 Rearrange Newton's equation to make force the subject. F = _____

7 **a** Assume that your mass is 50 kg. On earth, acceleration due to gravity is 10 m/s^2. Calculate your weight force, i.e. your weight. (Scientifically, weight is measured in newtons, not kilograms.)

_____

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Newton's laws of motion: applications 17b

b At an altitude of 1000 km above the earth's surface, gravity acceleration is only 7.3 m/s^2. Calculate your weight there. _____

c At an altitude of 10 000 km above the earth's surface, gravity acceleration is only 1.5 m/s^2. Find your weight there.

_____

d On the moon, gravity acceleration is 1.6 m/s^2. Find your weight on the moon.

_____

e On the sun, gravity acceleration is about 270 m/s^2. Determine your weight on the sun.

_____

NEWTON'S THIRD LAW: ACTION – REACTION Newton's third law is about pairs of forces acting on objects. An action force is acting on one object in one direction and an equal but opposite reaction force is acting on another object in the opposite direction. In many cases, one of the forces is not obvious and may be overlooked entirely.

8 **a** You are holding a book on your hand in front of you. The weight of the book is the action force. In the box at right, sketch your hand with the book on it.

b Draw an arrow from the centre of the book downwards. Label it the action force.

c State what is providing the balancing force, ensuring that the book is not moving.

_____

d State how big this balancing force is compared to the action force. _____

e Draw another arrow to represent this balancing force on the book. Label it the reaction force.

f Imagine a friend adds ten books to the first one. State how the action force has changed.

_____

g Explain what would have to happen to the reaction force to keep the books stationary.

_____

Another force pair involves the gravity force of the books on the earth and the gravity force of the earth acting on the books. These forces act on different objects.

h Which gravity force, the books' gravity force or the earth's gravity force, is bigger? _____

j State what would happen if the earth's gravity force were bigger than your arm muscle force.

_____

9 **a** The drive wheel of a car is pushing against the bitumen of the road and the car moves forwards. In the box at right, sketch a car wheel on a road surface.

b Draw an arrow starting at the wheel and pushing backwards, representing the action force. Label it.

c Draw another arrow as the ground is pushing against the wheel, representing the reaction force. Label it.

d State how those two forces are similar and different.

_____

_____

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Isaac Newton – Founder of classical physics 18a

Isaac Newton was born into a farming family in Woolsthorpe, England, in 1643. He went to the Free Grammar School in nearby Grantham. His school reports described him as idle and inattentive. Isaac mostly kept his own company and was known as a sober, silent, thinking lad. When he did associate with others, it was nearly always with girls. Isaac is remembered for his strange inventions and extraordinary inclination for mechanical works. Among these was a windmill powered by a treadmill run by a mouse.

Later, his mother made Isaac leave school in order to help run the family farm. But with support from his uncle, Isaac continued his studies at Trinity College, Cambridge, in 1661. Newton's aim at Cambridge was a law degree, but soon he became involved with other topics like algebra, optics and astronomy. Newton did a lot of studying of other scientists and ancient philosophers.

To test a question about colours, he stared at the sun with one eye until all the colours changed. As a result, he had to shut himself up in a dark room for several days before he could get rid of the spots floating before his eyes.

Despite some evidence that his progress had not been particularly good, Newton received his bachelor's degree in 1665. His scientific genius suddenly emerged when the plague closed the university in the summer of 1665 and he had to return home to Woolsthorpe. There, in a period of less than two years, he began revolutionary advances in mathematics, optics, physics and astronomy. In 1666 Newton had early versions of his three laws of motion. Newton had also reached the conclusion that white light is not a single entity, as most scientists since Aristotle had believed. When Newton passed a thin beam of sunlight through a triangular glass prism, the spectrum of the rainbow colours formed. However, Newton did not publish any of his work then.

When the University of Cambridge reopened in 1667 after the plague, Newton was elected to a major fellowship within a year.

Newton then became interested in alchemy. He laboured day and night in his chemical laboratory trying to produce gold from other metals. And then he immersed himself in mathematical and mystical calculations. In the late 1670s, theological studies occupied most of his time.

In 1686 Newton presented his single greatest work, the *Philosophiae Naturalis Principia Mathematica* ('Mathematical Principles of Natural Philosophy', simply known as the 'Principia').

In it, Newton revealed his laws of motion and the law of universal gravitation. The first law of motion was about inertia: a body wants to keep doing what it is doing. A moving body wants to continue moving in a straight line, while a stationary body wants to remain stationary.

The second law ($a = F / m$) said that the acceleration of an object depends on the force (stronger force means more acceleration) and its mass (bigger mass means less acceleration).

The third law stated every action force has an equal but opposite reaction force. Newton further demonstrated that all heavenly bodies mutually attract one another. Further generalisation led Newton to the law of universal gravitation. The astronomer Edmund Halley (of Halley's comet fame) paid for the publication of the Principia. The Principia redirected Newton's intellectual life away from theology and alchemy, back into science.

After suffering a second nervous breakdown in 1693, Newton retired from research and decided to leave Cambridge for a government position in London. He became the Warden of the Royal Mint in 1696 and Master in 1699. In 1703 he was elected president of the Royal Society and was re-elected each year until his death.

Newton was knighted by Queen Anne in 1705; the first scientist to be honoured in this way. His last decades were passed in revising his major works, polishing his studies of ancient history, and defending himself against critics, as well as carrying out his official duties. Newton was modest, diffident and a man of simple tastes. He was angered by criticism or opposition, and he harboured resentment; he was harsh towards enemies but generous to friends. He never married and lived modestly.

In 1709, Newton began work on a second edition of Principia and he also published a second edition of Opticks; however after he moved to London, he did nothing but reshuffle his old ideas. As his health began to deteriorate, he began to distribute his wealth amongst his family. He died in 1727 after a series of debilitating illnesses.

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Isaac Newton – Founder of classical physics 18b

Complete these tasks in your science book.

1 Make a timeline starting at 1640 and finishing in 1760. Mark and number every other decade above the line; mark and label Newton's age in increments of ten years below the line. Then indicate on the timeline the major events in Isaac Newton's life.

Reading on the lines ...

2 Why did Newton leave school?

3 What were Newton's interests at school?

4 School and university were not free in Newton's time. Who paid for his education?

5 What subject did Newton start studying at university?

6 Why did Newton return home after his university studies?

7 How long was there between Newton's graduation and his election to a major fellowship?

8 What is the short title for Newton's major book?

9 What is inertia?

10 What made Edmond Halley famous?

Reading between the lines ...

11 How old was Newton when he died?

12 How old was Newton when he started university?

13 Why would Newton have stared at the sun with only one eye?

14 What did an alchemist try to do?

15 How many years after conception of his ideas did Newton publish them?

16 Forces usually occur in pairs. What are their names?

17 What is the name given to the attraction between two masses?

18 Why did Newton leave university life?

19 How did Newton earn his money in London?

20 What exciting event happened to Newton in 1705?

21 Newton's scientifically creative period occurred during his two years in the country after he had graduated from university. Back in London, what scientific work did he do?

Reading beyond the lines ...

22 Why would Newton's mother have made him leave school?

23 What is wrong with staring at the sun for any length of time?

24 What is the scientific term for the bending of light in a prism?

25 What are the seven colours of the rainbow spectrum?

26 Why didn't Newton publish his ideas straight after conception?

27 What do the letters in $\mathbf{a = F/m}$ stand for?

28 How do you calculate the force needed to make an object move?

29 Why is Halley's comet named so?

30 Why would Newton have given his wealth to his relations?

31 Newton said that he stood 'on the shoulders of giants' when he developed his laws. What did he mean and who were these giants?

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Albert Einstein: E = mc²

Albert was born in Ulm in south Germany in 1879 and spent his youth in Munich, where his family owned a small shop that manufactured electric machinery. He did not talk until the age of three. As a youth he was curious about nature and able to understand difficult mathematical concepts. At the age of 12 he taught himself geometry.

However, Albert didn't enjoy school. After a break from school when he was 15, he sat for the entrance exam to a Swiss university. Sadly, he failed the Arts section and so he had to finish secondary school in Aarau, Switzerland. Finally, in 1896, he entered the Swiss National Polytechnic University in Zurich in order to become a science teacher. After Einstein graduated in 1900, he became a tutor. Einstein had also renounced his German citizenship in order to avoid military conscription and he adopted Swiss nationality. In 1902 Einstein could not get a position as a lecturer or teacher, so he took a job at the Swiss Patent Office. Although Einstein and Mileva Maric, another student, had a baby (who was adopted) in 1901, they only married in 1903.

In 1905, Einstein published five papers. The first (Generation and transformation of light) suggested that light consisted of quanta, bundles of energy which behaved sometimes like particles (e.g. in reflection) and other times like waves (e.g. in refraction). This contribution to quantum mechanics eventually earned Einstein a Nobel Prize in 1921.

The second paper (Motion of suspended particles and the kinetic theory) provided a mathematical explanation that molecules in liquids moved and a proof that atoms really did exist.

The most famous paper (on the electrodynamics of moving bodies) introduced the Special Theory of Relativity. This was the basis for a new physics: time, space and mass were all connected with each other, no longer separate and fixed, but different, relative to the observer. In a supplement of 1907 Einstein included the famous equation $E = mc^2$, which showed that matter and energy were interchangeable and no longer separate and constant. This formed the basis for nuclear reactors and bombs.

Because Einstein had no association with a university and because his ideas were difficult to understand, he was accepted only slowly by other scientists. Eventually, in 1909, his first academic appointment was at the University of Zurich. In 1911 he moved with his family to Prague, where he became a professor at the German University there. In 1912 he returned to the Swiss National Polytechnic in Zurich. And in 1913, he was appointed director of the Kaiser Wilhelm Institute for Physics in Berlin. Soon after, in 1914, Einstein and Maric divorced.

In the General Theory of Relativity (1917), Einstein explained the variations in the orbital motion of the planets and predicted the bending of light near a star. When the gravitational bending of light was observed during a solar eclipse in 1919, Einstein's fame spread worldwide. In this year Einstein also married Elsa Lowenthal, who had nursed him back to health two years before, when he had collapsed from exhaustion.

When Hitler came to power, Einstein decided to leave Germany for the United States in 1933. He took a position at the Institute for Advanced Study at Princeton, New Jersey.

In 1939 Einstein and several other physicists wrote a letter to President Roosevelt, pointing out the possibility of making an atomic bomb and the likelihood that the German government was working on it already. This letter helped to start the development of the atomic bomb, but Einstein himself played no role in the work and knew nothing about it at the time.

After World War II, Einstein was active in the cause of international disarmament and world government. He continued his active support of Zionism but declined the offer made by leaders of the state of Israel to become president of that country. Einstein died in Princeton in 1955.

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Albert Einstein: $E = mc^2$

19b

Complete these tasks in your science book.

1 Make a timeline starting at 1870 and finishing in 1960. Mark and number every other decade above the line. Mark and label Einstein's age in increments of ten years below the line. Then indicate on the timeline major events in Albert Einstein's life.

Reading on the lines ...

2 In which country was Einstein born?

3 What did Einstein teach himself during his early high school years?

4 In which country did Einstein finish his schooling?

5 Why couldn't Einstein enter university?

6 What job was Einstein interested in after university?

7 Einstein again couldn't secure a desired position. What job did he take up instead?

8 Why did Einstein renounce his German citizenship?

9 What is a quantum (plural: quanta)?

10 Why didn't other scientists accept Einstein's theory straight after publication?

11 What was Einstein's letter to Roosevelt about?

Reading between the lines ...

12 Why would Einstein have taught himself geometry?

13 Why did Einstein study in Switzerland and not Germany?

14 How many years after his paper did Einstein receive a Nobel Prize?

15 Newton defined space and time. In what way did his concepts differ from those proposed by Einstein?

16 According to Einstein, what has to happen to space if time changes?

17 How many years after publishing his famous papers did Einstein begin working in a university?

18 Why was Einstein suddenly famous during 1919?

19 Why did Einstein leave Germany when Hitler came to power, even though he was a Swiss citizen?

20 Einstein was a pacifist. How did he show this in his life?

21 How old was Einstein when he died?

Reading beyond the lines ...

22 Why may Einstein have disliked school?

23 When Einstein was 15 years old, his family lived in Milan, Italy. Why didn't he attend school there?

24 How does light behave like a particle during reflection?

25 What is refraction?

26 How do molecules in liquids move?

27 What is an atom and what is it made of?

28 Who had defined space time and mass before Einstein? How much earlier?

29 What do the letters in the expression $E = mc^2$ represent?

30 The development of the atomic bomb in the USA was known as the Manhattan Project. What role did Einstein play in it?

31 Why would Einstein have been unhappy about his contribution to the development of nuclear energy?

32 Why would Einstein have been invited to become the president of Israel?

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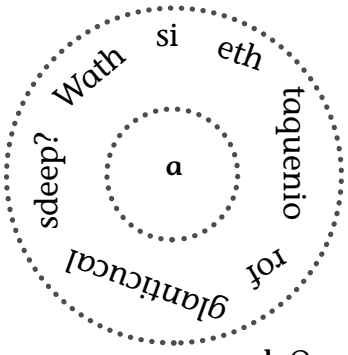
Wheels of motion

1 Unjumble each word in the wheels. Write the question, then answer the question.

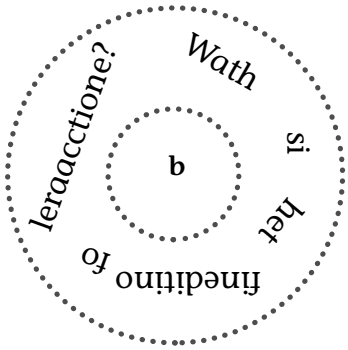


a Question _____

_____

_____

Answer _____

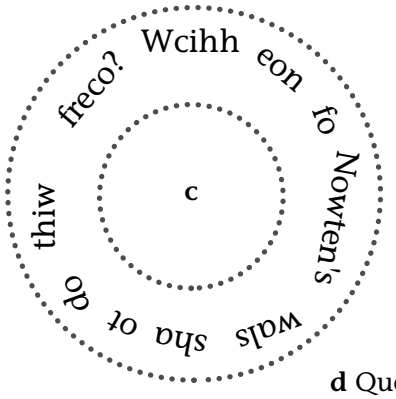


b Question _____

_____

_____

Answer _____

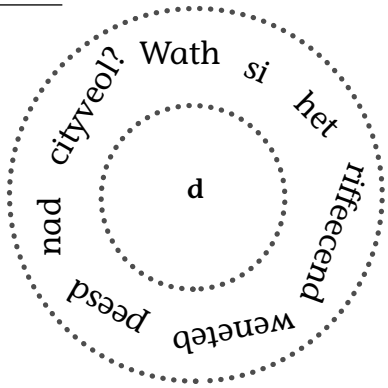


c Question _____

_____

_____

Answer _____



d Question _____

_____

_____

Answer _____

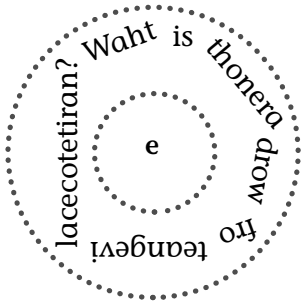
e Question _____

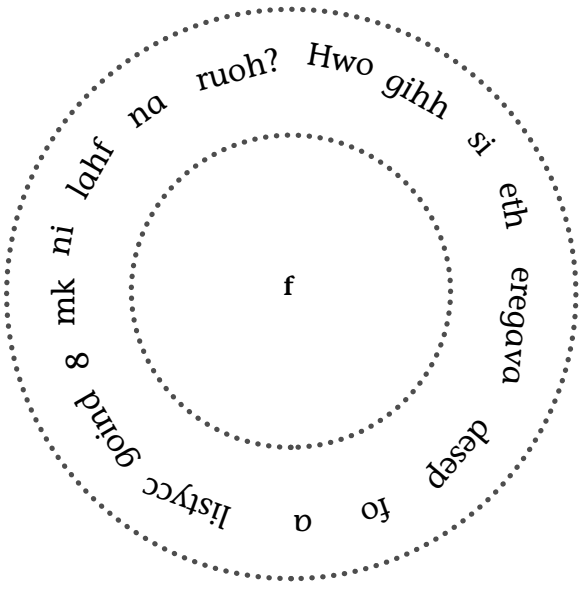
_____

_____

_____

Answer _____





f Question _____

_____

_____

Answer _____

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Answers

1: MEASURING SPEEDS – A 50-METRE TRACK

Group results	Walking (s)	Jogging (s)	Running (s)
Average time calculation	(33.3 + 32.0 + 35.4 + 28.3 + 30.0)/5	(22.2 + 25.3 + 23.0 + 26.4 + 24.1)/5	(7.5 + 7.0 + 8.0 + 9.5 + 8.5)/5
Average time answer	31.8 s	24.2 s	8.1 s
Average speed calculation	50/31.8	50/24.2	50/8.1
Average speed (m/s) answer	1.6 m/s	2.0 m/s	6.2 m/s
Average speed (km/h)	6 km/h	7 km/h	22 km/h

2: HOW DO SPEEDS COMPARE?

1 0 m/s; 36 km/h; 22 m/s; 4 km/h; 17 km/s; 19 km/h

2 11 m/s; 22 km/h; 30 m/s; 7 km/h; 6 m/s; 72 km/h

3 277.8 m/s; 1080 km/h; 27.8 m/s; 1 080 000 000 km/h; 103.3 m/s; 5 400 km/h

4 0 km/h, 36 km/h, 72 km/h, 108 km/h, 144 km/h, 180 km/h, 288 km/h

3: FORMULAS FOR ENERGY AND MOTION

2 **a** $a = (v_2 - v_1)/t = (100 - 0)/7 = 14.3$ km/h/s OR $a = (v_2 - v_1)/t = (28 - 0)/7 = 4$ m/s^2
b KE = $\frac{1}{2}$ m v^2 = $\frac{1}{2}$. 1000 . 28^2 = 392 000 J = 392 kJ; 98 kJ; 24.5 kJ; 0 J. **c** 140 m **d** 7.1 s **e** 2 MJ
f –7 m/s^2 (decelerates: 7 m/s^2) **g** –56 m/s^2

3 **a** 13 kJ. **b** 600 m **c** 50 s **d** –5 m/s^2 **e** 325 N
f 20 m (about seven storeys)

4: THE CONVENIENT UNIT

1 Metre (m); second (s); kilogram (kg); litre (L); newton (N); joule (J); ampere (A); volt (V); byte (b)

2 **a** milli, 1/1000, m; micro, 10^{-6}, μ; nano, 1/1 000 000 000, n; pico, 10^{-12}; kilo, 1000, k; mega, 10^6; giga, 1 000 000 000, G
b micrometre - 1 μm - 10^{-6} m; nanometre - 1 nm - 10^{-9} m; kilometre - 1 km - 10^3 m; megametre - 1 Mm -10^{12} m; gigametre - 1 Gm - 10^9 m

3 100 m; 40 km; 1 mm; 0.1 mm (100 μm); 300 Mm; 150 Gm; 50 μm; 0.2 μm (200 pm)

4 **a** 1 ms - 0.001 s; 1 μs -1 000 000 μs; 60 s - 1/60 min 1 h - 3 600 s - 1/3 600 h; 1 day - 86 400 s - 1/86 400 day; 1 y - 31 536 000 s - 1/31 536 000 y.
b 8 min; 17 min; 360 min (6 h); 30 min (0.5 h); 1 s; 10 ps; 7.2 s; 36 s

5: HOW LONG WILL IT TAKE?

1 2 m/s = 2 . 1/1000 km /1/3600 h = 2 . 3600/1000 km/h = 2 . 3.6 km/h = 7.2 km/h.

2 250 m/s; 6 944.4 m/s; 27.8 m/s; 8.3 m/s; 13.9 m/s; 41.7 m/s; 22.2 m/s; 1.4 m/s; 2.8 m/s; 5.6 m/s; 36 km/h; 6.9 m/s; 11.1 m/s

3 0.1 km; 0.5 km; 2 300 m; 1 000 000 m; 300 000 m.

4 Speed = distance/time (v = d/t)

5 Time = distance/speed (t = d/v)

6: SPEED AND THINGS

1 Objects remain stationary or continue to travel, when no unbalanced force acts on them.

2 When the car stops, inertia means passengers will continue to travel. Seat belts apply force to stop motion.

3 The bus stopped but the passenger kept moving because of his/her inertia.

5 **a** 1 and 3 s; 9 and 10 s **b** 6 and 9 s **c** 3 and 6 s.

6 **a** $= (v_2 - v_1)/t = (24 - 6)/12 = 1.5$ m/s^2

7 **a** $v_1 = 25$ m/s **b** $v_2 = 0$ m/s **c** –5 m/s^2
d Negative: deceleration

8 **a** Graph 1; Graph 2; Graph 3

9 $t = (v_2 - v_1)/a = 3$ s

10 Velocity is speed with direction

11 Speed = distance/time

13 60 km/h

14 **a** 16.7 m/s **b** 27.8 m/s **c** 720 km/h

15 10 km/h; 50 m/s; 10 km/h; 1000 m/min; 120 km; 6000 m; 3 s; 10 h

16 Speed-time graph **b** 4 – 7 s, 9 – 11 s, 12 – 14 s, **c** 6 m/s **d** 9 s **e** Time (s): 0, 4, 7, 9, 11, 12, 14, 20; Speed (m/s): 0, 3, 3, 5, 5, 6, 6, 0.

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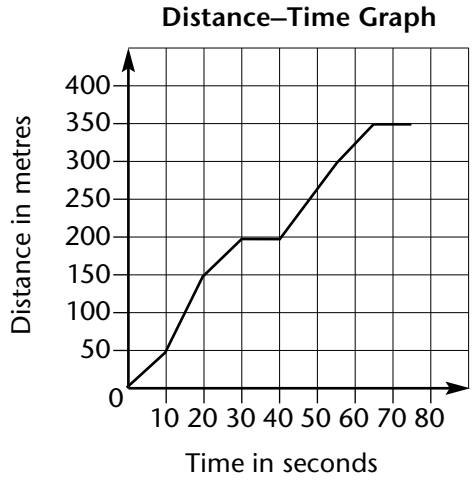
Answers

7: CAR CONTROL

1



The column entries for time (s) are 0, 10, 20, 30, 40, 55, 65, 75 and the corresponding entries for distance (m) are 0, 50, 150, 200, 200, 300, 350 and 350.

2 **a** Car stationary **b** Moving forward **c** High speed **d** Going backwards **e** D and G; When it is zero; B; B

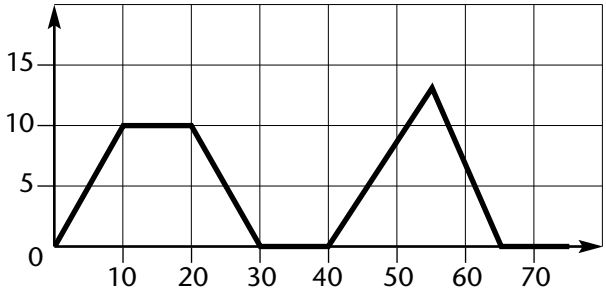
3 **a** Speed = distance / time

b

Section	t_1 (s)	t_2 (s)	t (s)	d_1 (m)	d_2 (m)	d (m)
A	0	10	10	0	50	50
B	10	20	10	50	150	100
C	20	30	10	150	200	50
D	30	40	10	200	200	0
E	40	55	15	200	300	100
F	55	65	10	300	350	50
G	65	75	10	350	350	0

	d/t (m/s)	Speed (m/s)	Final speed v
A	50/10	5	10
B	100/10	10	10
C	50/10	5	0
D	0/10	0	0
E	100/15	20/3 = 6.7	40/3 = 13.3
F	50/10	5	0
G	0	0	0

c



4 **a** Travelling at constant speed **b** Increasing speed or accelerating **c** Accelerating most quickly **d** Accelerating more slowly **e** Slowing down or decelerating **f** Decelerating most quickly **h** zero – B, D, G; lowest – E; highest – A; smallest – C; greatest – F

5 **a** Acceleration = change in speed/time

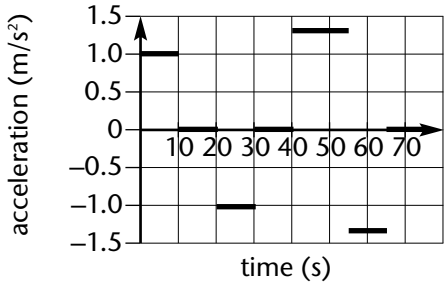
$a = (v_2 - v_1)/t$

b

Section	t_1 (s)	t_2 (s)	$t_2 - t_1$ (s)	v_1 (m/s)
A	0	10	10	0
B	10	20	10	10
C	20	30	10	10
D	30	40	10	0
E	40	55	15	0
F	55	65	10	13.3
G	65	7.5	10	0

v_2 (m/s)	v (m/s)	Speed/Time	Average acceleration (m/s^2)
10	10	10/10	1
10	0	0/10	0
0	–10	–10/10	–1
0	0	0/10	0
13.3	13.3	13.3/10	1.3
0	–13.3	–13.3/10	–1.3
0	0	0/10	0

c



6 **a** Accelerating or increasing its speed **b** Driving at constant speed **c** Decelerating or decreasing its speed

8: GRAPH STORIES

1 **a** Kim **b** Sam **c** Mia **d** Jane **e** Chris

9: SPEED AND VELOCITY – WHAT'S THE DIFFERENCE?

1 mean, thing, there, two; travels, rate, km/h, its say, is travelling, km/h, south, are talking; specify, and, are, velocity; and, are; exists, is, has been covered, states, is; is, is; tells, is, is; is, and; is; is calculated; is found.

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Answers

2 **a** Constant speed and constant velocity also are – not the same. Constant velocity describes an object – moving at constant speed in the same direction; – for example, a car driving along a long straight – path. However, a car can move at a constant speed – along a curved path like a roundabout. But – this car does not have a constant velocity, because – its direction is changing continually in the roundabout.
b Sometimes average speed and instantaneous speed are confused. Average – speed is found by dividing the travelled distance – by the time it took to cover that distance. Instantaneous speed – is the speed at which an object is travelling at a particular time. For – example, imagine you were cycling to a friend's house. It took you – one hour to cover the ten kilometres. So your average speed is 10 – km/h. But when you stopped at the – lights, your instantaneous speed was 0 km/h, and when you raced – downhill, your instantaneous speed might well have been 30 km/h.

3 Displacement: s, m west, vt; distance: d, m - vt; speed: v, m/s, d/t; time: t, s, s/v; velocity; v, m/s east, s/t

4 **a** the amount of ground travelled **b** shortest distance between start and finish **c** how fast an object is travelling **d** how fast you are going in a certain direction **e** the total distance travelled in unit time **f** the speed at a particular moment.

6 30 km/h; –; –, 50 m/s E; 3 h, 40 km/h S; 5 s; –; 250 km N, 50 km/h; 800 m W, 20 m/s

10: TRAVELLING FAST

3 Suggestions include: kilometres per hour, km/h; kilometres per second, km/s; metres per second, m/s; metres per hour, m/h

4 **a** 3.6 m/s **b** 1 km/h = 1/3.6 m/s = 0.28 m/s **c** 1.39 m/s
(5 km/h); (6.67 m/s), 24 km/h; (5 m/s), 18 km/h; (0.83 m/s), 3 km/h; 6.94 m/ s (25 km/h)

5 **c** 0.005 h (18 s); 0.03 h (108 s); 0.007 h (25.2 s); 2.7 h

6 **c** 50 km/h: 13.89 m/s, 27.78 m, 7 cars – 80 km/h: 22.22 m/s, 44.44 m, 11cars – 110 km/h, 30.56 m/s, 61.11 m, 15 cars

11: STOPPING REALLY FAST

1 Faster stopping features: Wide tyres, Driver only, Small car, Disc brakes, Combined disc/ drum brakes, Big brake pads, ABS brakes, Alert driver, Low speed, Racing car driver, Smooth bitumen road, Dry road.

5 20 km/h – 5.6 m; 40 km/h – 11.1 m; 50 km/h – 13.9 m; 60 km/h – 16.7 m; 80 km/h – 22.2 m; 110 km/h – 30.6 m; 200 km/h – 55.6 m

6 Speed (km/h): 20, 40, 50, 60, 80, 100, 110
Braking distance (m), 3, 10, 15, 22, 38, 60, 72.

7 Speed (km/h), 20 – Reaction distance (m), 5.6 – Braking distance (m), 3 – Stopping distance (m), 8.6 etc.

9 Faster, more, more.

12: TICKER TAPES AND OIL DROPS

1 The distance between the dots increases.

3, 4, 6

Gap number	1	2	3	4	5	6	7	8	9	10	11
Gap distance (mm)	2	4	6	8	10	12	14	16	18	20	22
Gap time (s)	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
Average speed (mm/s)	20	40	60	80	100	120	140	160	180	200	220
Average speed (mm/s)	20	40	60	80	100	120	140	160	180	200	220

8 **c** The slope of the smooth line is the speed.
d speed = (2 + 4 + ... 22 = 132)/1.1 mm/s = 12 mm/s

10 **a** For the first gap the acceleration is 20/0.1 mm/s^2 = 200 mm/s^2 **b** The acceleration is constant.

13: CARS AND GEARS

1 160 km/h

2 37 s

3 First gear

4 First gear

9 16.7 km/h/s; 7.5 km/h/s; 5.0 km/h/s; 2.7 km/h/s; 1.5 km/h/s

12 **b** a = (160 – 0)/37 = 4.3 km/h/s

13 **a** The curve is upwards **b** downwards
c horizontal

14: BABY ON BOARD

1 F = ma

2 –20 km/h

3 5.56 m/s

5 –22.2 m/s^2

6 133.2 N

7 **c** 13.3 kg

8 55.56 ms^{-2} acting on 6 kg gives force of 333.3 N: 33.3 kg

9 53.3 kg

10 122.2 kg

11 50 kg: 263.9 kg. 70 kg: 369.4 kg

12 12 500 kg

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Answers

15: MASS AND WEIGHT – WHAT IS WHICH?

1 **a** 0 N **b** 60 kg **c** 0 kg **d** 0 N

2 **a** 60kg **b** 10kg **c** 100 N **d** 6 m

3 **a** 300 times bigger **b** 60 kg
c 60 . 300 = 18 000 kg
d 30 000 / 300 = 100 g = 0.1 kg

4 300 000 times bigger **b** 60 kg
c 60 . 300 000 = 18 000 000 kg = 18 Mg
d 30 000 / 300 000 = 0.1 mm

5 **a** 50 . 1 = 50 N **b** 5 kg

6 **a** 50 . 150 = 7 500 N **b** 750 kg

16: NEWTON'S LAWS OF MOTION: HISTORY

1 He was knighted by the British king

2 He was insecure or uncertain

3 1686 in London

4 Edmond Halley

5 motion

6 inertia

7 keep moving

8 stay still

9 keep moving in a straight line

10 keep falling

11 a = F / m

12 acceleration also increases

13 acceleration decreases

14 action force, reaction force

15 backwards

16 also 5 N

17 12 N downwards

18 **a** yes **b** theoretically yes, but only an infinitesimal distance

17: NEWTON'S LAWS OF MOTION: APPLICATIONS

2 **a** Seatbelts restrain forward and sideward motion, especially in a crash. **b** Riders need to be clear of the bike in a crash. **c** City buses don't travel as fast; commuters stay on city buses for shorter times/ distances. **d** So travellers stay in their safe seats as the plane goes up/ down suddenly. **e** The lift moves away from under you and you press down less on the floor.

3 m/s^2; N kg

4 a = F / m = 7500/1500 = 5 m/s^2

5 a = F / m = 200/100 = 2 m/s^2

6 F= m a

7 **a** F = m a = 50 . 10 = 500 N **b** F = m a = 50 . 7.3 = 365 N **c** F = m a = 50 . 1.5 = 75 N **d** F = m a = 50 . 1.6 = 80 N **e** F = m a = 50 . 270 = 13 500 N = 13.5 kN

8 **c** hand pushing up **d** equal size **f** action force is eleven times bigger **g** hand has to push up eleven times harder **h** neither, both are equal but opposite **i** books fall down as hand is pulled downwards

9 **d** same size; opposite direction, acting on different objects

20: WHEELS OF MOTION

a What is the equation for calculating speed? Answer: v = d / t

b What is the definition of acceleration? Answer: change in speed in a certain time

c Which one of Newton's laws has to do with force? Answer: second law, F = m a

d What is the difference between speed and velocity? Answer: velocity is size and direction; speed has no direction, only size

e What is another word for negative acceleration? Answer: deceleration

f How high is the average speed of a cyclist doing 8 km in half an hour? Answer: 8 / 0.5 = 16 km/h

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